

Review article

Feature selection for online streaming high-dimensional data: A state-of-the-art review

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ABSTRACT

Knowledge discovery for data streaming requires online feature selection to reduce the complexity of real-world datasets and significantly improve the learning process. This is achieved by selecting highly relevant subsets and minimising irrelevant and redundant features. However, researchers have difficulties in addressing various forms of data. The goal of this article is to present a state-of-the-art review of feature subset selection based on the data form for the high-dimensional data used in online streaming. Through a systematic literature review assessing journal and conference papers from the past five years, detailed discussions on traditional feature selection and online feature selection were presented. Subsequently, a taxonomy of the challenges related to OFS provides a comprehensive review of state-of-the-art OFS and the benchmark methods. Several data forms were identified based on the extensive review: group stream, multi-label, capricious, imbalance, and feature drift. Using critical analysis, the evaluation metrics of online feature selection methods were compared from the perspectives of threshold initialisation, accuracy, high dimensionality, running time, relevancy, and redundancy for the optimal feature subset. An online feature selection framework was derived to illustrate the relationship between the application area, data form, online feature selection methods, evaluation metrics, and tools. Finally, the findings and potential directions for future research were thoroughly discussed. It is suggested that future researchers explore the derived framework and aim to advance each method.

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1. Introduction

The unprecedented scale of data that has emerged from different sources, such as the internet of things, social media, bioinformatics, and other data resources, has significantly impacted areas of research into Big Data [1]. The emerging 7Vs of Big Data – variety, velocity, volume, veracity, variability, visualisation, and value [2,3] – require extensive research in terms of knowledge discovery and patterns. These aspects have been the subjects of specific research into the various layers of big data problems. Feature selection is one of the problem areas of big data that involves high dimensionality. Feature selection is a knowledge discovery tool used to provide insights into a problem by interpreting relevant features while handling the aspects of dimensionality reduction [4].

A significant phase of knowledge discovery and database (KDD) is pre-processing. In feature selection, this phase involves the discovery of optimal features, including (1) feature selection to find relevant features and (2) feature extraction to apply transformation methods [5]. Hence, feature selection has been firmly defined as finding a subset of features according to given criteria by selecting relevant features and removing non-informative features, such as those that are irrelevant or redundant [6]. Unlike the feature extraction method, feature selection retains the value of data without any alteration, despite reducing the dimensionality [7,8]. The significant goals of feature selection are to elevate the performance of a model through prediction accuracy [9], improve model interpretability [10], and develop a robust model against overfitting to enable better generalisability [11].

Since machine learning can benefit from feature selection methods, which reduce the model complexity and training time [12], extensive studies have been conducted to apply feature selection in machine learning tasks. These include classification [13–15], regression [16–18], and clustering [19,20]. Surveys were conducted on the previous literature related to feature selection data behaviours. The first major issue discovered was the struggle to deal with the volume of data due to the lack of

scalability [21–23]. This issue correlates with feature selection because not all features have the same amount of discriminative power and some contain redundant and irrelevant data [24,25]. Increasing the volume of data leads to emerging feature space, especially with high-dimensional data [26]. This is known as the curse of dimensionality, which denotes the exponential growth of volume due to the increment in the number of dimensions or features [27]. Hence, researchers have focused intensively and narrowly on traditional feature selection methods such as the filter [28,29], wrapper [30–32], embedded [33–35] and hybrid methods [36–38] to enhance the performance of feature selection in high-dimensional data. This is applicable to static data and most research continues to focus on managing static datasets within vast volumes of data [39]. However, the recent behaviour of big data has become so complex that traditional feature selection cannot effectively process it [12].

The other significant issue discovered in relation to big data was velocity. Researchers have broadened the subject by experimenting with feature selection across different data environments. This has led to further developments in feature selection needs, especially for data stream environments [40–43]. There are three major aspects of learning and extracting knowledge from continuously flowing data: feature selection, concept drift, and high-dimensional data [44–47]. Research works in these areas demonstrate limitations in terms of the batch mode of feature selection, while selecting relevant features in a timely manner has been complicated [23,48]. Since feature selection cannot wait for the arrival of all the features, Online Feature Selection (OFS) has been designed to deal with streaming data. Nevertheless, most OFS techniques that have been claimed to be applicable in streaming data might fail because features need to be discovered in advance [15,49]. Although the goal of OFS is to select highly relevant features with class labels and minimise redundancy among selected features [50], determining the optimal feature subset remains a complex challenge because of the continuous feature arrival and high-dimensional feature spaces [22].

Furthermore, some surveys that included discussions on OFS have been reviewed [21,22,40,51]. AlNuami et al. [40] reviewed

Table 1
Summary of the gaps in the existing surveys.

Year	Ref.	Gap
2017	[21]	OFS methods were not covered in depth in this survey and there was a research gap in terms of data behaviours in OFS.
2019	[22]	The authors only reviewed some benchmark OFS methods, disregarding one of the most important OFS categories — Online Group Feature Selection (OGFS) — and neglecting to undertake a performance evaluation of OFS.
2019	[40]	Qualitative studies of OFS methods in terms of performance evaluation are still lacking, based on this review.
2022	[51]	The study still failed to examine some aspects of OFS, like data behaviours and OFS tools performances — or conduct a comprehensive evaluation of performance measurement.

the batch mode feature selection methods and scrutinised the OFS methods proposed by past researchers by presenting the strengths and weaknesses of each method in meeting the challenge of handling a big data stream. However, qualitative studies of OFS methods in terms of performance evaluation are still lacking, according to the review. Meanwhile, Rong et al. [22] approached their review of feature selection methods from two main perspectives: the different characteristics of big data and the different application categories. These applications are text classification, image classification, and microarray analysis. As the review aimed to examine the topic from the perspective of big data, the authors only reviewed a selection of benchmark OFS methods, disregarding one of the most important OFS categories, Online Group Feature Selection (OGFS), and neglecting to conduct a performance evaluation of OFS. Li et al. [21] systematically evaluated recent advances in feature selection research from the perspective of behaviours of different forms of big data, including heterogeneous, structured, conventional, and streaming data. However, OFS methods were not covered in depth in this survey, and data behaviours in OFS was a research gap in this work. Abdulwahab et al. [51] presented a comprehensive review of the state-of-the-art feature selection methods in the context of big data, organised into a structured taxonomy that categorised feature selection methods based on the search strategy, nature, and performance. Despite a thorough survey of the state-of-the-art and several experiments, the study still failed to examine some aspects of OFS — like data behaviours and OFS tools performances — or undertake a comprehensive evaluation of performance measurement.

Based on the gaps in these surveys listed in Table 1, the current research extended the topic by focusing on OFS to comprehensively investigate OFS data behaviours, correlation measures, and evaluation metrics, as well as the tools used particularly for OFS and to deal with various OFS issues. As shown in Table 2, this was achieved by, firstly, providing a well-defined taxonomy of the challenges that complicate OFS tasks and procedures. This taxonomy will assist future researchers attempting to capture complex data streams so that substantial research can be undertaken to develop more advanced OFS methods. This was the first objective of the current study, as shown in Fig. 1, and will contribute extensively to research that handles more complex data stream types.

Hence, based on the taxonomy, this review derived a classification framework to provide ideas for addressing the challenges accordingly. This was the second objective of this study, as shown in Fig. 1. These objectives were fulfilled using critical and rigorous methods to explore the various state-of-the-art reviews. The justifications for and contributions of these objectives are shown in Table 2.

This article is organised into several sections. Section 2 provides a taxonomy of the OFS challenges in dealing with complex high-dimensional data streams. Section 3 presents the research methodology employed in analysing the data. In this study, a

Systematic Literature Review (SLR) was employed to investigate thoroughly feature selection, streaming data, high-dimensional data, and OFS. Section 4 presents the derived OFS classification framework. This section also discusses in depth each component of the framework, namely (1) An overview of the feature selection taxonomy that categorised feature selection in terms of handling static and streaming data, (2) Data sources and domains that contribute to (OSH) data, (3) Benchmark OFS methods, (4) OFS classification methods based on data forms and subset evaluation, (5) Analysis of the evaluation metrics and (6) OFS tools. Section 5 contextualises the findings of the state-of-the-art studies by linking the challenges and the derived framework. Finally, the conclusion and suggestions for future work are presented in Section 6.

2. Online feature selection and challenges

Big data analytics is entering a new stage known as the fast data phase, in which large amounts of data (in terabytes or more) arrive in big data systems every second. The 7Vs — volume, value, velocity, variety, variability, visualisation and veracity — are the fundamental components in modern big data systems that capture complex data streams.

The process of learning and extracting knowledge from the continuous flow of data is defined as data stream mining. The number of features used for an evaluation determines the efficiency of the data stream mining process. However, dealing with high-dimensional data with relevant and redundant features complicates feature selection [52]. Hence, streaming feature selection, known as the Online Feature Selection (OFS) method, is needed to accommodate this streaming data behaviour. To deal with streaming features, OFS utilises the stream mode by reassessing the existing features with the newly received features in a single [15,22] or group stream [53–55]. In terms of energy consumption and computational cost, collecting reduced and relevant streams of data is deemed to be more valuable than collecting raw, inconsistent, redundant, and noisy data. Current research into the volume, velocity, and variability is increasing exponentially and is pertinent to the future of handling high-dimensional data. However, additional studies are required to further enhance the development of feature selection in handling more complicated data stream types. The following taxonomy of OFS challenges identified in this paper are shown in Fig. 2. On this basis, further comprehensive research can be undertaken in this field.

2.1. Data behaviour

Data behaviour will always affect the performance of feature selection since data behaves differently and must be treated accordingly. Some data behaviours like concept drift and multi-label are real-world behaviours and still need to be studied more extensively.

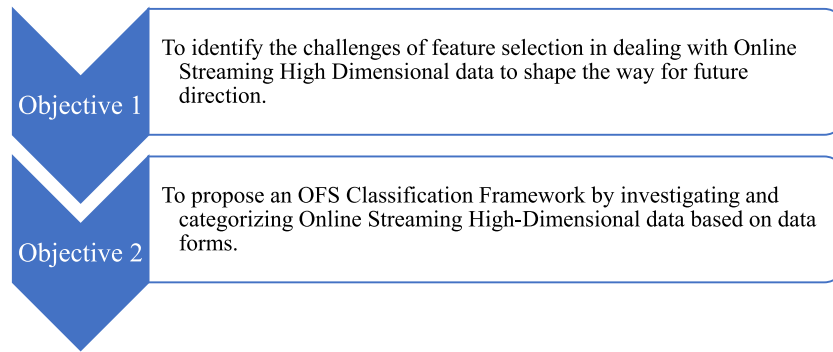


Fig. 1. Objectives of the study.

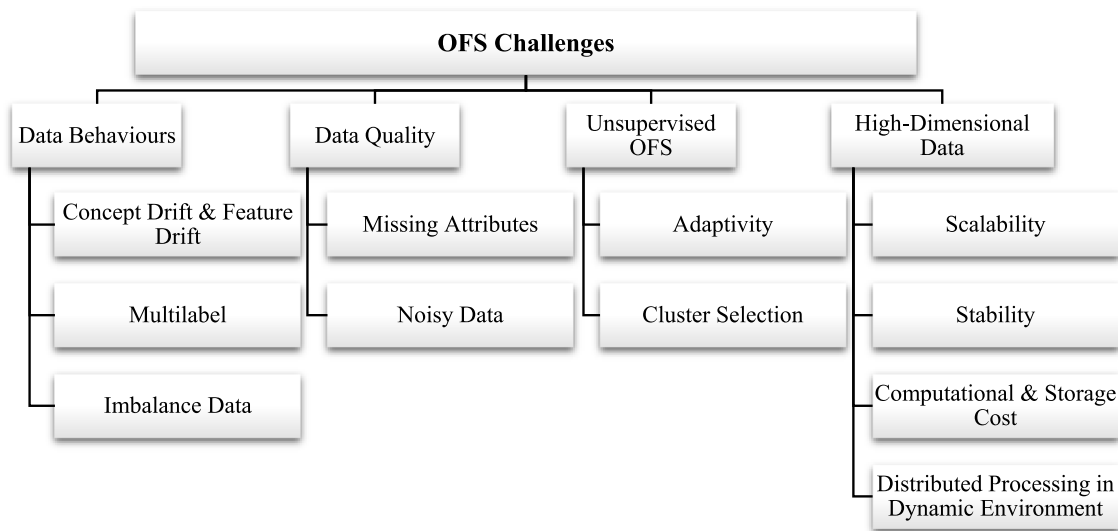


Fig. 2. Taxonomy of OFS challenges.

Table 2
Justifications and Contributions.

	Justifications	Contributions
1	Providing a well-defined taxonomy of the challenges that complicate OFS tasks and procedures.	i. The taxonomy will assist future researchers by capturing complex data streams so that extensive work can be undertaken to develop more advanced OFS methods.
2	Deriving a classification framework to provide ideas for addressing the challenges, based on the defined taxonomy.	i. A review of feature selection taxonomy, focusing on feature selection with streaming high-dimensional data obtained from various domains and data sources. ii. A comprehensive survey and comparative analysis of the state-of-the-art papers related to OFS, highlighting their contributions and limitations and categorised according to the data forms (data behaviours) and correlation measures for subset evaluation. iii. A qualitative analysis of the performance of the reviewed OFS methods based on different evaluation metrics derived from threshold initialisation, data dimensionality, as well as relevancy and redundancy state adjustments. iv. An evaluation of the capabilities of the various tools used, particularly for OFS methods with different data behaviours.

2.1.1. Concept drift and feature drift

OFS is considered batch learning since when the number of features is large, the number of instances remains constant and does not change over time. Additionally, the inclusion of features in an OFS model is often performed sequentially [56], but not all feature streams are in single and group sequence. Hence, OFS does not consider actual behaviour for all data streams. The concept of drift behaviour becomes more complicated as the importance of features changes dynamically over time. Important features can become irrelevant, and vice versa, depending on the concept learnt [57]. Specifically regarding the behaviour of

data, data streams are prone to various forms of drift over time, including changes to the relevance of feature value, which forces the learner to adapt its predictive model. Concept drift behaviours require investigation in terms of both features and instances as it is considered another branch of OFS that has not been explored in depth [58,59].

Generally, this is addressed as a feature drift issue, in which the data stream is prone to change in terms of feature characteristics or in which disparities exist in feature importance, such that a feature that has been important may later become meaningless, or vice versa [60]. One common example of data

with concept drift and feature drift behaviour is text streams [61]. Feature space in text stream problems consists of unique terms that are potentially unbounded, resulting in the relevancy and redundancy of feature changes over time. For this reason, recent advancements in the field have emerged in terms of online learning, but the evolution of OFS remains unaligned with online learning [62,63]. In data streams, incrementally identifying essential attributes is critical in this research area. Therefore, new approaches must be developed to allow the feature selection process to detect and adapt to changes in feature relevancy. To date, insufficient studies have been conducted on OFS while also handling the impact of feature drift.

2.1.2. Multi-label

In recent years, multi-label categorisation of static data has gained substantial interest [64–66]. However, because static data solutions require static concepts and the availability of all data for learning, they do not easily transition to stream settings [67]. A multi-label data stream has numerous targets (concepts), and it is unrealistic to expect them all to start drifting at the same time or rate [50]. Essentially, each concept is expected to observe its own drift pattern, so the feature subset selection will continue to change over time [68,69]. Handling OFS for multi-label data remains a challenging issue that requires further exploration.

2.1.3. Imbalanced data

Machine learning and data mining communities from both academia and industry have recently expressed interest in learning from imbalanced datasets. Instead of over-sampling or under-sampling the training samples, feature selection uses a different approach to tackle the imbalanced class problem. The fundamental idea is to find a subset of features that best corrects the disparity between classes in the dataset; the best features are then chosen to represent both classes [70]. Zhou et al. [71] proposed that a future direction could be to use OFS on multi-class imbalanced data and on other real-world applications with high class imbalanced data. Real-world applications with high class imbalance include fraud detection, cancer detection [72], and text mining in the clinical domain [73], which require additional studies, especially in regard to handling streaming data.

2.2. Data quality

Data quality cannot be guaranteed in real-world applications due to issues such as missing attribute values and noisy data. As a result, selecting high-quality features from a feature stream is difficult [74]. Data collected by networked sensors and communication equipment is often used in streaming analytics [45]. For example, if a sensor in a medical monitoring system generates incorrect data, streaming analytics may transmit a false signal to the controlling devices, which eventually would potentially lead to patient deaths [75]. The pre-processing phase, including feature selection, is illogical when the data quality control process has not been established. Incomplete data is an interesting but formidable issue in data mining [76]. Feature selection allows a data model to be reconstructed in line with the original dataset but with much missing information due to the efficiency with which it explores the relationship between features and features with class attributes. The way that reliable methods of feature selection could be applied in a static environment to deal with noisy or incomplete data in streaming and dynamic environments remains an open issue. Moreover, feature selection with big data in a streaming and dynamic environment is highly problematic due to its large scale or high dimension [22].

2.3. Unsupervised OFS

Streaming data also has an impact on the unsupervised method. Since streaming data is continuous and rapidly changing, the underlying cluster structure is also affected and varies. To capture the streaming data changes and keep the optimal features updated, handling feature selection via unsupervised methods requires an algorithm that comprehends the behaviour of adaptivity and single-scan clusters [29]. Adaptivity is defined as an unsupervised algorithm; it should be able to deal with feature weights by readjusting and reselecting the set of new features while handling concept and feature drift [20]. Meanwhile, single-scan means the algorithm could cluster the incoming stream in a single scan. Streaming data or data that cannot be loaded into memory necessitate a single data scan since the second pass is either unavailable or prohibitively expensive. Moreover, selecting the number of clusters, K is an unresolved issue since limited, if any, knowledge is available on the domains in the streaming environment [77]. Limited attempts have been made to find the optimal parameters desirable for good clustering quality in online feature selection, which still requires considerable attention from researchers.

2.4. High-dimensional data

In this section, four main challenges were identified and further scrutinised to better understand high-dimensional data in handling OFS with streaming features.

2.4.1. Scalability

Scalability is defined as the influence of an increased training set size on the computational efficiency of an algorithm in terms of accuracy, training time, and allocated memory [78]. Due to the exposure of big data, research works still utilising traditional approaches are now straining to cope with the extremely high dimensionality of big data as they strive to extract good results in a fair amount of time. Big data, especially streaming data that is exceedingly multidimensional, cannot be loaded into memory in a single data scan. As a result, obtaining a feature relevance score without considering the appropriate density surrounding each sample is difficult. There are two distinct phases of the various methodologies for large-scale feature selection. The first assesses the relevancy of individual features before ranking them based on their relevance scores. In the second phase, only values with the highest relevance are used as input. However, this technique has the drawback of potentially removing low-ranking characteristics or even considering their relationships with other features [40].

The scalability of unsupervised feature selection algorithms is also a major issue. These algorithms usually necessitate a large number of samples to produce statistically acceptable results. It is difficult to judge feature importance without accommodating the density of each sample. Some approaches have attempted to solve this problem by memorising only the most important samples. Hence, the scalability of clustering and feature selection methods should correspondingly be prioritised to match data's rapid growth and fast streaming.

2.4.2. Stability

Stability is defined as the sensitivity of the feature selection process to data perturbation in the training set; hence, it measures the effect of the training set on feature selection. The performance of the feature selection algorithm is evaluated using classification accuracy. As a result, when constructing feature selection algorithms, the stability of any algorithm is important. Conversely, in terms of streaming feature selection, unknown features are the main challenge. The stability of any proposed

algorithm is challenged by selecting the most informative and relevant features from among the current features. Consequently, altering the chosen subset challenges an algorithm's robustness. The issue of stability is often dependent on the data dimensionality (m), sample size (n) and data distribution across the fold [79]. Developing feature selection algorithms for streaming data with high classification accuracy and stability in high dimensions remains an open challenge.

2.4.3. Computational cost and storage cost

Because data is frequently not static and is generated in a stream, feature selection techniques for both feature and data streams are frequently used in practical applications. Most existing algorithms in this area use various procedures to accelerate the selection process so that it can cope with new data samples or features as soon as they arrive.

Pre-processing approaches with low computational cost need to be developed, which would enable real-time decision making when working with large and high-speed data streams. With data stream mining in feature selection, it is important to balance the accuracy achieved, features selected, and computational time cost. For this reason, developing approximation solutions with stopping criteria, especially in circumstances of abrupt changes of stream data, would be beneficial.

Furthermore, most of these algorithms require numerous data passes, and some even demand the storage of all previously generated data, which makes them impractical to use when memory or disc storage is restricted. Further work is needed to develop effective and efficient streaming algorithms with low storage costs.

2.4.4. Distributed processing in dynamic environments

Centralised online feature selection algorithms will progressively struggle to fulfil the demand for computing speed as the amount of data grows. A dynamic environment occurs when the concept can change continuously and drift the data distribution [80,81]. Under dynamic circumstances, despite the availability of various streamwise models, the streaming data speed is a major obstacle. In some circumstances, it is more important to employ a simple, basic model than a precise one, as the former can describe the essential aspects of the original model. Parallel processing approaches have been used to make mining big data more efficient. In feature selection, data can be distributed in parallel horizontally or vertically before being combined into a single subset of relevant features [82]. Although feature selection approaches can make it quicker to create a data mining model, the benefits of both feature selection and distributed online parallel processing for streaming data should be further explored. Future data can be processed roughly online using the obtained data model, followed by other offline processing techniques. Hence, distributed online feature selection algorithms will be a new future challenge.

The challenges of OFS are taxonomised in this section. In addition, future directions are presented, each linked to every challenge respectively to pave the way for extensive future research. To address the challenges, an OFS classification framework had to be proposed by reviewing OFS from different perspectives and then linking the challenges cohesively. The findings of this study will serve as guidelines for newcomers to help them realise the challenges, while future OFS studies may be based on our proposed OFS framework classification.

3. Research methodology

The Systematic Literature Review (SLR) method was selected for this research to review feature selection, streaming data, and high-dimensional data. Since the SLR employs systematic searching strategies, proper guidelines are in place for conducting a rigorous review to obtain the relevant, high-quality, and latest research papers, as well as compiling a comprehensive analysis of the existing literature reviews. The three significant SLR phases adopted were planning, conducting the review, and interpretation [83]. In the planning phase, two research questions were posited as the benchmark for starting the research. The rationales for examining each research question led to the objectives. The research questions and objectives are shown in Table 3.

In the same planning phase, the relevant papers were selected in two stages. First, systematic searching strategies were conducted using credible scientific databases, namely Scopus, Web of Science, SpringerLink, IEEE Xplore (IEEE), ACM Digital Library (ACM), and Google Scholar. Then, using the snowball searching strategy, a comprehensive study was conducted based on the reference list retrieved from the systematic searches for papers. Table 4 reports the distribution of the relevant topics, based on the research questions, from the different database sources: Scopus, Web of Science, SpringerLink, IEEE Xplore (IEEE), and ACM Digital Library (ACM).

The second SLR phase was the review. This phase involves a systematic searching procedure based on criteria identification, using precise keywords or well-defined search strings to aid the retrieval of relevant papers. Several search strings were utilised to locate pertinent articles from the previous five years related to OFS, relevancy, redundancy, irrelevancy, and high-dimensional data. The following search string was used:

TITLE-ABS-KEY ("feature selection" AND "online feature selection" OR "stream feature selection" OR "online streaming feature selection" OR "high*dimension*") AND ("redundan*" OR "*relevan*") AND (LIMIT-TO (PUBYEAR, 2022) OR LIMIT-TO (PUBYEAR, 2020) OR LIMIT-TO (PUBYEAR, 2019) OR LIMIT-TO (PUBYEAR, 2018) OR LIMIT-TO (PUBYEAR, 2017))*

The screening or filtering process was also carried out throughout the systematic searching stage. This procedure is followed to determine the inclusion or exclusion of papers based on certain criteria. The inclusion criteria were as follows:

1. Research works on state-of-the-art methods published in the previous five years (from 2017 to 2022)
2. Studies in the English language
3. Studies related to feature selection
4. Research works on online feature selection or streaming feature selection
5. Research works on feature selection in high-dimensional data

The eligibility and relevancy processes followed the screening stage. The eligibility process ensures that all the articles selected during the screening process fulfilled the required criteria. In this process, the titles, abstracts, and keywords were reviewed to determine if each paper met the inclusion criteria. In the relevancy phase, each article was comprehensively reviewed, classified using different criteria based on the keywords, and evaluated using the NVIVO tool for refined analysis. The classified research papers were analysed based on the topic of the data behaviour of streaming data.

Table 3
Research question and research objective.

	Research questions	Research objectives
RQ1	What are the existing challenges of feature selection and online feature selection?	To identify the challenges of feature selection in dealing with online streaming high-dimensional data to shape the way for future direction.
RQ2	How does online streaming data behaviour affect online feature selection?	To propose an OFS classification framework by investigating and categorising online streaming high-dimensional data based on data forms.

Table 4
Number of research papers found from different database sources.

	Number of papers	Percentage
Scopus	54	21%
WOS	76	29%
Springer Link	78	30%
IEEE	43	17%
ACM	7	3%
	258	100%

4. Classification framework of online feature selection

A proper OFS classification framework is essential in providing insights into OFS from different perspectives while providing ideas to effectively manage the challenges that had been taxonomised, as shown in Fig. 2. Research works were reviewed to contextualise insights into OFS and potentially also enlighten future researchers. The classification framework for this research comprised (1) feature selection taxonomy, (2) data source and domain area, (3) OSH data behaviours, (4) subset evaluation, (5) evaluation metrics, and (6) tools. Since the focus was on handling feature selection for streaming data, the data was defined in this study as Online Streaming High-Dimensional data (OSH data) and the reviewed articles were classified based on the behaviours of OSH data. The OSH data behaviour categories employed were group stream, high-dimensional, multi-label, class imbalance, capricious, causal-based, low redundancy, high relevancy, and feature drift. Based on these categories, it was determined the subset evaluation should include rough set theory, information theory-based techniques, Markov Blanket, Bayesian Network, symmetrical uncertainty, the evolutionary algorithm, multi-objective optimisation, windowing, and regularisation.

Next, because most previous surveys had not focused intensively on OFS evaluation measures, the intention of this research was to further investigate the impact of threshold initialisation, dimensionality, accuracy, running time and relevancy, and redundancy analysis. It was hoped to provide solutions to some of the existing and identified challenges. Finally, it was hoped that the framework would enable another component to be deduced, OFS tools, to explore how tools affected different data behaviours, while it was also hoped to explore HDD behaviour further. The classification framework depicted in Fig. 5 was designed to aid researchers in properly addressing various OFS behaviours and effectively dealing with other OFS challenges.

4.1. Feature selection taxonomy

The predictive ability of a learner gradually decreases with the growth of feature dimensionality. Machine learning becomes particularly complex when the number of features is enormous but the number of samples is low. The search space will be sparsely populated, making the model unable to differentiate between relevant data and noise. A new computational problem will arise with large sample sizes and a high dimensionality of data [84]. Getting more data could theoretically enable more discriminating ability. However, in some cases, a high dimensionality of data will lead to the “Curse of Dimensionality”, also

known as “the Hughes effect” [16,85]. Feature selection methods are alternative approaches employed by machine learning to minimise dimensionality [86,87]. Unlike feature extraction, feature selection produces results that keep the selected features in their original shape [88]. The use of a feature selection method is a recognised data pre-processing phase used for data reduction, which detects relevant features and removes irrelevant and redundant features to obtain a small optimal subset of features [89].

Feature selection significantly improves learning performance, leads to better accuracy and model interpretability with minimal deterioration, and reduces the computational time [90]. Fig. 3 illustrates the segregation between feature relevance and feature redundancy. It consists of four subsets:

- (I) Irrelevant Features
- (II) Weakly Relevant and Redundant Features
- (III) Weakly Relevant but Non-Redundant Features
- (IV) Strongly Relevant Features.

Subsets (III) and (IV) will be determined as the optimal subset. Redundancy is usually defined in multivariate cases to yield a compact subset of features, whereas relevance is established for individual features [16,91]. Further definitions of feature relevancy, redundancy, and optimal subsets are presented in the following section.

Definition 1 (Feature Relevance). It can be intuitively determined that a feature is relevant if it contains some information about the target. Relevance was initially defined by John et al. [92], who suggested that it comprised two degrees: weak and strong. Later, Kohavi and John [93] classified features more formally into three distinct categories: strongly relevant, weakly relevant, and irrelevant features. The definition of each category is given below: *Strongly Relevant*: A feature X is strongly relevant if the removal of X results in the model accuracy decreasing.

Weakly Relevant: A feature X is weakly relevant if it is not strongly relevant and, if it exists in a subset of features S , the performance of the prediction model deteriorates.

Irrelevant: A feature X is defined as irrelevant if it gives no useful information in the final prediction model.

Definition 2 (Feature Redundancy). Feature redundancy refers to the degree of interdependence among features. When two features are redundant, they are non-independent and completely correlated in terms of information. Consecutively, a redundant feature needs to be removed from the subset if there is weak relevancy and a Markov Blanket within the subset [94]. The redundancy between features is typically measured using Approximate Markov Blanket, Mutual Information, correlation, and Symmetrical Uncertainty of Features.

Definition 3 (Optimal Subset). Let a dataset be denoted by features, $F = [f_1, f_2, \dots, f_d]$ from a distribution, P over the labelled instance space and prediction model, L . An optimal feature subset, f_{opt} , is defined as a subset of the features that will obtain the best accuracy in the prediction model. Originally, the optimal subset

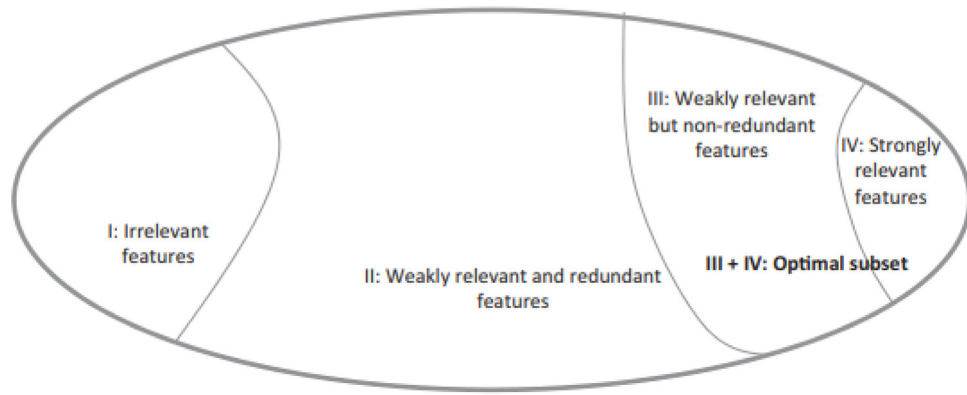


Fig. 3. Feature relevance and redundancy segregation [40].

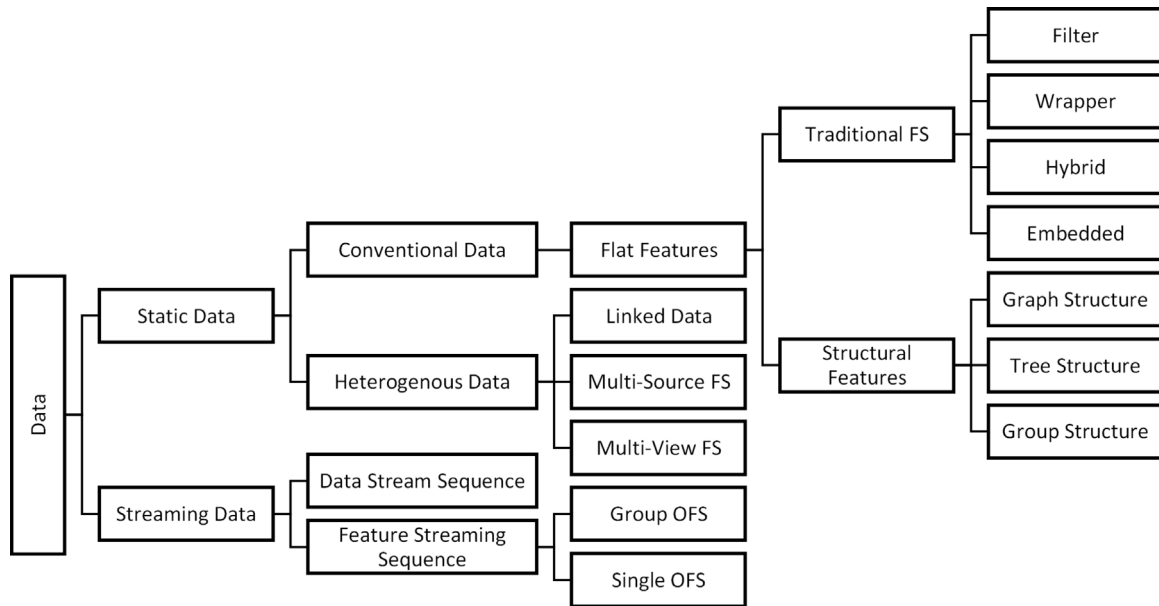


Fig. 4. Feature selection taxonomy [21,31,40].

was defined as the best subset that a feature subset generation can select with the accuracy of the induced classifier being at the maximal [93,95].

Since the three main Vs of big data are volume, velocity, and variety, efficient feature selection methods need to be reconsidered [96]. Feature selection in big data presents both substantial challenges and opportunities due to the characteristics of the data. According to the research by [21], feature selection was reviewed based on the data perspective, which led to a critical review of feature selection for static data, streaming data, structured data, and heterogeneous data. Fig. 4 shows the feature selection taxonomy from a data perspective, aggregated based on research by [23,31,40]. The next explanation refers in detail to the classification framework depicted in Fig. 5.

4.1.1. Traditional feature selection with static data

Several feature selection methods have been proposed for high-dimensional data. In traditional feature selection, five categories of feature selection methods can be used to analyse the relationship between the evaluation measure and the classifier. These are filter, wrapper, embedded, ensemble, and hybrid methods. The filter method is an independent measure from the classifier. Thus, this method benefits the feature subset evaluation, leads to an efficient, lower computational cost, and enhances the generalisation ability [11,85]. One drawback of this

method is that it is not useful when no interaction occurs with the classifiers [26]. However, filter methods can be beneficial when combined with others. Several researchers have applied filter methods to handle handwriting recognition, binary medical datasets, and other domains [97–99]. Wrapper methods use a subset of features to train a model by capturing feature dependencies [100]. From the generated model, features are added and removed from the subset selection based on the existing interaction with the classifiers in this method [101]. Problems occur when it is classifier-dependent since there will be a risk of overfitting and it can be computationally expensive [102].

Practically any search strategy and modelling algorithm combination can be employed as a wrapper. Still, wrapper methods are only feasible for greedy search strategies [36] and fast modelling algorithms such as Naïve Bayes, linear SVM, and Extreme Learning Machines [34,103]. In wrapper methods, a learning algorithm is applied to iteratively assess the quality of feature subsets in the search space [104], so this involves a high computing cost for high-dimensional datasets.

The embedded method resulted from the benefits of the filter and wrapper methods [72,105]. Promisingly, the embedded method deals with learning algorithms at a lower computational cost than the wrapper method [31,34]. It also captures the dependencies between the features and the interaction with the

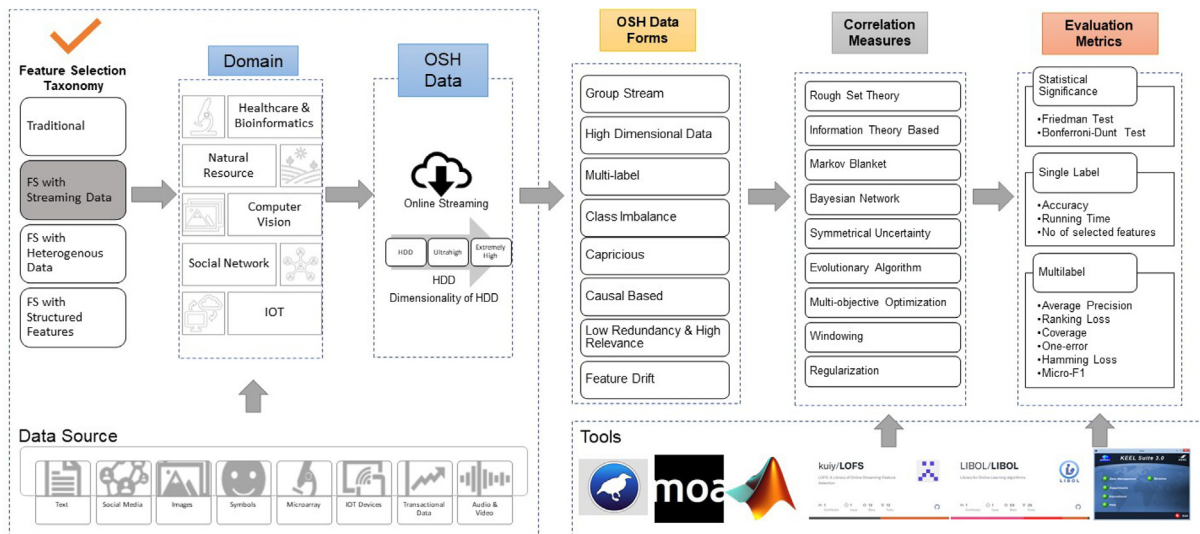


Fig. 5. Classification framework of online feature selection.

classifiers since it performs feature selection during the training process based on certain classifiers. Obviously, searching for an optimal subset using the embedded method occurs in classifier construction [26].

Due to the advantages of both the filter and wrapper methods, hybrid feature selection methods can be highly efficient and yield desirable accuracy and speed results when dealing with high-dimensional data [106]. Hybrid methods have been used in both supervised [38,107] and unsupervised feature selection [37] to efficiently overcome the challenges of search space and high-dimensionality. Meanwhile, ensemble methods have been examined as a viable framework to strengthen the robustness of the selection process, particularly in high-dimensional, small sample size scenarios. By utilising numerous models, ensemble methods have enhanced the extraction of stable features in a wide range of applications [108,109].

The available feature selection approaches are based on a strong presumption that features are mutually independent (flat), ignoring feature structures. However, features may have various structures in several forms of actual real-world implementation, including spatial or temporal smoothness [23], group structure [110], trees, and graphs [111]. Feature selection with group feature structures assumes that features are generated individually with group structures, a common example of which is groups of gene sets in biology [112]. Feature selection with tree features will simulate the hierarchical tree by generating root and subtree values for hierarchical classification [15,113]. Meanwhile, in graph structure feature selection, strong pairwise correlation data – such as natural language processing and gene studies – requires analysing and preserving the correlation between datasets [21]. Meanwhile, heterogeneous data from different sources requires feature selection methods that can deal with issues of noise, redundancy, and the partial structure of data that is to be merged [25]. Previous researchers have investigated feature selection that deals with heterogeneous data in the linked data [25], multi-source [114,115], and multi-view [116,117] contexts.

4.1.2. Feature selection with streaming data: Online feature selection (OFS) framework

The previous methods have assumed that all the data is available in advance. However, in real-case situations, data, especially that from the IoT and social media, exists in streams called streaming data. One general example of streaming data is online

spam email detection [23,118,119], in which new emails arrive constantly. Subsequently, it is not easy to appropriately handle streaming data in a timely manner using batch mode or traditional feature selection. A clear distinction between data streams and feature streams is needed in the definition of streaming data. Data streams exist when the instances change over time with fixed features [120–122], while feature streams deal with dynamic features with static instances [40]. When the feature space is unknown or infinite, it is preferable for OFS to be employed in streaming data [123]. Another reason why OFS has attracted attention is that feature space, although known, requires an exhaustive search of the available and incoming data, making it infeasible with large feature space [124].

Motivated by the major challenges facing OFS, feature relevancy, and redundancy, a general framework was needed to investigate the underlying OFS mechanism. Some researchers have discussed the need to review the notation of relevancy, as well as adopt redundancy and irrelevancy for online feature selection due to the nature of streaming features that change over time [125,126].

Fig. 6 illustrates the general flow steps of OFS, as follows:

1. Populate new features (single/group stream).
2. Determine the addition of new features to the selected subset by relevancy, redundancy, and irrelevancy analysis.
3. Update the subset of the existing features.
4. Repeat steps 1 to 3 until all the feature space has been examined and the optimal subset has been found.

Diverse implementation is undertaken using different methods. Some studies ignore the redundancy process and only handle relevant and irrelevant data [127,128]. Meanwhile, not all methods consider the best candidate subset, taking into consideration a comparison of the previous subset features as an optimal subset, as in Step 3 [129,130]. Since this article focused on OFS, further explanation is provided in the following sections of the benchmark OFS methods, the survey of the state-of-the-art OFS methods classified through different data behaviour, and the performance evaluation of the OFS methods.

4.2. Online streaming high-dimensional data (OSH data): Domain and data source

OSH data exists in different domains such as healthcare and bio-informatics [131,132], natural resource management [123,

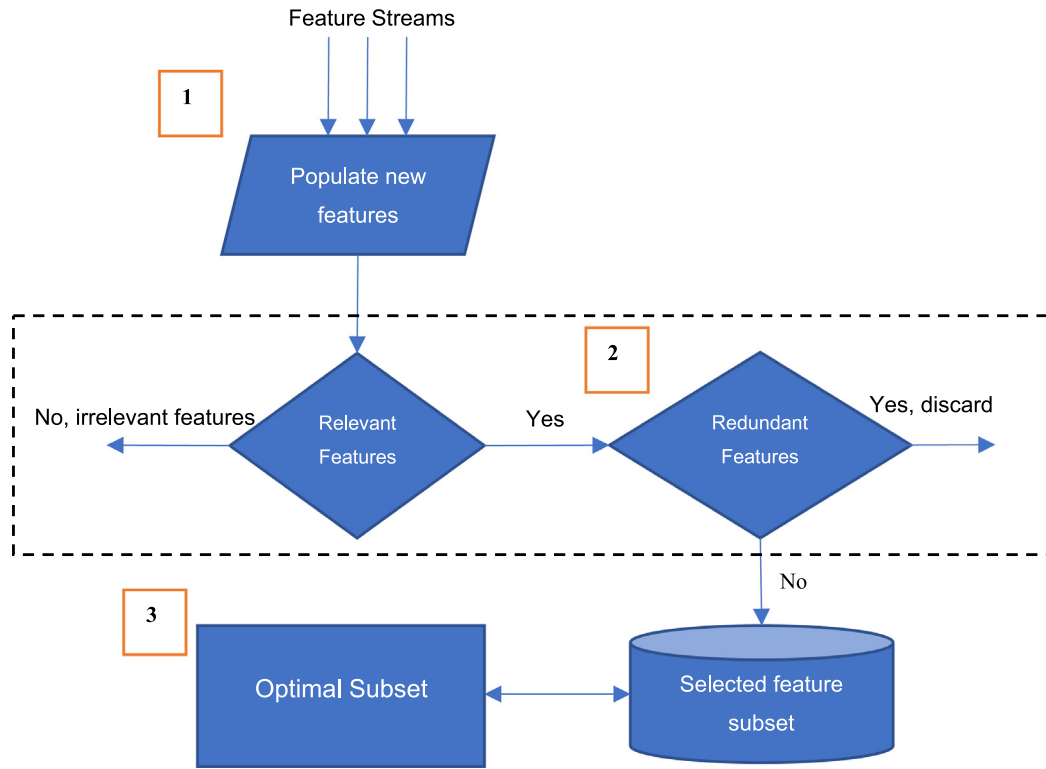


Fig. 6. General flow of OFS.

133,134], computer vision [135], social networks [136,137] and the IoT [63,138]. These domains generate data into different sources that include text [50,61,136,139], social media [17,75], images and symbols [55,140], microarrays [26], IoT devices [141,142], transactional data [143,144], as well as audio and waves [34,34].

4.3. Benchmark OFS method

Online feature selection methods are mainly divided into two types, online single-stream feature selection and online group feature selection. Both types have been proposed to solve problems based on feature stream characteristics and both have proven effective and efficient in regard to streaming features.

4.3.1. Single-stream feature selection

Researchers first developed various basic OFS approaches such as Grafting and Alpha-Investing [53], followed by those who addressed other gaps and challenges. In this section, the benchmarks and novel methods are analysed, and the advantages and disadvantages of each method are explored, as shown in Table 5. Grafting can discard irrelevant and redundant features using a stagewise gradient descent and optimised with a regularised framework. However, Grafting requires global feature space for regularisation parameters in advance, a major drawback when handling streaming features. Besides, the enormous dimension of features makes the Grafting runtime substantially longer [145]. Another drawback of this method is that when features selected in the optimal subset are later found to be redundant, Grafting cannot discard them [129].

Alpha-Investing, known as streamwise feature selection, is a popular algorithm for analysis and it has the advantages of ignoring the determination of prior parameters and being able to handle vast feature sets [146]. Regardless of its capabilities, one major drawback of this method is not considering redundancy

for an optimal subset. It also fails to handle features in a sparse dataset [147].

By considering the problems affecting Grafting and Alpha-Investing, researchers have developed another novel OFS method, Online Streaming Feature Selection (OSFS) [148]. OSFS is divided into three evaluation categories: irrelevance, weak relevance, and strong relevance, while two significant OSFS analyses have been established: online relevance analysis and online redundancy analysis. The former adds strongly or weakly relevant features to the optimal subset and simultaneously discards irrelevant features. Meanwhile, the latter identifies redundant features in a dynamic way. The process of redundant identification is believed to be able to add new further features to the global selected features. Hence, OSFS is suitable for high-redundancy feature streaming and sparse candidate feature sets [149]. The OSFS drawback is the redundancy aspect, which is known to be inefficient, especially with large feature space. The same researcher later extended the study to address the OSFS problem by developing an advanced algorithm called FAST-OSFS. Using this algorithm, the efficiency was improved significantly by dividing the redundancy process into two phases to handle unknown and larger feature space using Markov Blankets. Nevertheless, OSFS and Fast-OSFS faced another problem when dealing with group feature streams [150,151].

SAOLA, which stands for the Scalable and Accurate Online Approach, is another recent OFS benchmark that addresses the challenges of high dimensionality and group features [152]. SAOLA adapts the pairwise correlation technique with the current feature subset and class attributes. SAOLA also employs the k-greedy search for the optimal subset by searching for redundant features, and it can handle group streaming features, thus providing a better approach than OSFS and Fast-OSFS. The drawback of SAOLA is that it requires an optimal value for the relevance threshold. OFS-Density and OFS-A3M are two new methods, recently developed by Zhou et al. [124,153], that have received attention from

Table 5
General characteristics of benchmark OFS methods.

Methods	Advantages	Disadvantages	Requires prior knowledge	Threshold specification	Time used as benchmark	Ref.
Grafting	Can handle large instances. Can discard many irrelevant and redundant features and have a single global optimal solution.	Requires global info. Weak in streaming data. Time-consuming when the features are very large.	Yes	No	2	[154]
Alpha-Investing	Dynamically adjusts the threshold for adding features to the model, so it is conducive to reducing overfitting.	No redundancy measure. Not efficient for huge streaming features.	Yes	Yes	14	[146]
OSFS	Increased accuracy for most datasets. Allows incremental updates.	Time-consuming. Ignores feature interaction.	No	Yes	14	[148]
SAOLA	Can handle large data.	Needs threshold specification. Hard to get optimal value for relevance threshold.	No	Yes	11	[155]
Fast-OSFS	Has a memory for redundant features.	Scalability issue.	No	Yes	12	[148]
OFS-A3M	Runs faster. Can select features with high correlation, high dependency, and low redundancy.	Selects more features when more high-redundancy data is used.	No	No	5	[153]
OFS-Density	Selects features with low redundancy.	The final selected feature subset may differ, based on whether the training instances are different. Not faster than OFS-A3M or SAOLA.	No	No	4	[124]

other researchers. This is because these two methods can employ non-thresholding feature selection methods to obtain the optimal subset. OFS-Density has the advantage of being able to handle low-redundancy datasets; nevertheless, it is not as fast as SAOLA in running time. Motivated by the latter issue, OFS-A3M was developed and, it has been claimed, can run faster while selecting features with high correlation, high dependency and low redundancy. However, it is not efficient when the datasets have more redundant features available in feature sets as it will select more features. Further analysis of the state-of-the-art OFS-Density and OFS-A3M is presented in the following section.

4.3.2. Group stream feature selection

Handling group feature selection and single feature selection are different as the features arrive by groups in the former. Group feature selection was introduced by GFSSF [126] and was aimed at differentiating group and individual features using information theory techniques. Extensive studies were undertaken, subsequently leading to the creation of OGFS [156] and group-SAOLA [155]. The OGFS algorithm was introduced for image classification and face recognition; good results were obtained on a number of selected features by using Lasso to handle redundancy. However, OGFS has an exponential runtime that causes the method to explode with more non-redundant features [157] and a parameter is required in advance to select an optimal value. When handling big data, Group-SAOLA, which was introduced as an extended version of SAOLA, outperformed other group feature selection methods by using fewer resources such as memory and it could also handle sparse data [155].

4.4. Online feature selection methods classification based on data form and subset evaluation: State-of-the-art

In the recent literature, researchers have tackled different OFS challenges, including unknown feature space, the optimisation of thresholding-based methods, and handling different forms of data. In this section, the state-of-the-art methods are classified based on data behaviour. These forms include group stream, high-dimensional data, multi-label data, class-imbalanced data, capricious data, low-redundancy and high-relevance data, feature drift, structured data, and noisy data. Certain correlation measures are required to handle some forms of data. However, some researchers have hybridised different correlation measures to elevate the performance of their proposed OFS methods. These data

forms are mapped to their particular correlation measures, such as rough set theory (classical rough set, neighbourhood rough set, fuzzy rough set), information theory (mutual info), Bayesian network, and symmetrical uncertainty. Some OFS methods choose to optimise the basic approaches by hybridising windowing, multi-objective, and regularisation methods. Table 6 present the classification of the data forms with the correlation measures of recent OFS methods.

4.4.1. Online feature selection on group stream

SFGS was proposed for the group feature stream by embedding new incremental techniques that assigned new incoming features to existing groups or created a new group. SFGS measures the symmetrical uncertainty (SU) search strategy with regard to the class to determine the value of each feature. By using feature grouping and nomination of centroid, the author determined redundancy in relation to the existing features after adopting the relevant features [53]. However, SFGS still requires a user-defined parameter to control the distance threshold when building a new cluster or adding a new feature to an existing one.

By considering the feature interaction in group feature selection, OGSFS-SI was developed based on a hybridisation of Mutual Information Theory, Symmetrical Uncertainty, and Markov Blankets with an elastic net [55]. Based on intra-group and intergroup selection, it was claimed that OGSFS-SI could efficiently select relevant and interactive features while filtering out redundant ones.

In the intra-group component, Mutual Information (MI) was used to measure the interaction of more than two variables and extended using the Multi-Information method. MI has a drawback when selecting features with more values, so Symmetrical Uncertainty was adopted to normalise the MI measures by calculating the information among feature subsets with decision features. OGSFS-SI further divided weakly relevant features into non-redundant and redundant features using the Markov Blanket algorithm. The intergroup selection component of this method adopts an elastic net algorithm to find the global optimal subset through the regularisation of the elastic net threshold.

4.4.2. Online feature selection with high-dimensional data (HDD)

Recently, researchers have experimented with handling high-dimensional data (HDD) when proposing OFS methods. Based on the analysis of state-of-the-art OFS papers dating from 2019 to 2021, some researchers have started to employ higher dimensions of datasets, using up to millions of features in their

Table 6

Classification of Data Forms with the Correlation Measure of State-of-the-art OFS methods.

Data form/correlation measure	Classical rough set	Neighbourhood rough set	Fuzzy rough set	Markov blanket	Information theory	Symmetrical uncertainty	Windowing	Multi-objective	Evolutionary algorithm	Bayesian network	Regularisation
Group stream		SF-DSW-FRL [158]			OGSFS-GI [55]	SGFS [53]	SF-DSW-FRL [158]	G-SFS [136]	G-SFS [136]		
High-dimensional data		OFS-density [124] OFS-A3M [153]	DFSR [142]		SF-DSW-FRL [158] OFSFMI [137] OSMWFS [159]	OMGFS [41] ACO-SU [135]		MOANOFs [42]	LFBA [160] ACO-SU [135] OFS-ABA [133]	BS-SFS [118]	BS-SFS [118]
Multi-label		OM-NRS [161] OFSI [123] SF-DSW-FR [158]			FSSL [113] MSFS [162] SF-DSW-FRL [158]	OMGFS [41] Pareto-FS [163]	SF-DSW-FR [158]	I-SFS & G-SFS [136] Pareto-FS [163]			
Class Imbalance		K-OFS [164] OFS-KW [131]									MOFS [165]
Capricious				GF-CSF [166] OFSFW [147] OMBSF [52] OFS-MR SDMB [87] CondInd	CD-SU-SF [132]	CSSU [167] CD-SU-SF [132]					
Causal-based											
Sparse data		OFS-KW [131]							LFBA [160]		
Low redundancy				CondInd [149] OFSFSCM [130]	OFSFSCM [130]						
high relevancy											
Feature Drift											OFSER [61]
Noisy data	OS-NRRSAR-SA [129] OFS-MRMS [150]				OS-NRRSAR-SA [129]						
Structured data				SDMB [87]						SDMB [87]	

experiments, as depicted in Fig. 7. In 2021, the highest dimension of the datasets used was 20Newsgroup, with a dimensionality of up to 1,355,191 features [42], which could be considered ultra-high-dimensional. However, in the past three years, not a single research work has been able to cater for datasets of up to nearly 30 million dimensions, as SAOLA had done in 2014 [152]. Different search strategies that applied OFS methods to handle HDD are discussed in the following section.

Classical Rough Sets were originally proposed to deal with categorical data since some real-world application datasets cannot be reliably handled. Numerous studies have applied neighbourhood rough sets to deal with discrete, numerical, and heterogeneous data [123,124,131,153,165]. Neighbourhood rough sets use the neighbourhood relation to replace approximation, based on the equivalence relation [168]. Motivated by the challenge of specifying parameter values in advance when using neighbourhood rough sets in traditional feature selection methods, Zhou et al. [153] defined OFS-A3M as an online stream feature selection method that was constructed based on a new neighbourhood rough set and adaptive neighbourhood relationship. Using the maximum dependency, maximum relevancy, and maximum significance evaluation criteria, it is capable of selecting features with high dependency, high correlation, and low redundancy with high-dimensional data. However, OFS-A3M tends to select more redundant features due to its capability to select more features.

In another study by the same author, OFS-Density [124] was designed based on the adaptive neighbourhood relation by applying the density information of the surrounding instances, which selected features with a low redundancy using fuzzy equal constraint. For a very sparse dataset, OFS-Density was shown to select fewer features [123]. The parameter λ , used for fuzzy equal constraint, makes OFS-Density consider more features for the redundancy analysis. The drawback of OFS-Density is its inconsistent selection of the final selected feature subset when the training instances are different. Furthermore, it is no faster than OFS-A3M and SAOLA in terms of running time and cannot handle hierarchical class structures [15].

Another representative extension of the rough set is the fuzzy rough set, which models vagueness to accommodate the uncertainty of the classical rough set model [99]. Moreover, it can feasibly handle numerical and fuzzy data [112]. The distributed fuzzy rough set technique has been utilised in a distributed fuzzy rough set (DFRS) based on feature selection and was specifically applied in cloud computing [142]. Using incremental learning, the DFRS can obtain a reduction with bare redundancy, significantly overcome computer resource constraints, and apply a fuzzy rough set model on enormously large datasets. This would subsequently enable more meaningful information to be extracted. However, on datasets with multiple decision classes, the algorithm produces a high memory proportion and runtime. The DFRS demonstrated low accuracy on the Yeast and Glass dataset, exhibiting multiple decision class behaviours with accuracy values of 0.3 and 0.6, respectively.

OSFSMI is an OFS method that implements Information Theory using Mutual Information (MI) in both relevancy and redundancy analyses to achieve linear computational time – $O(nm)$. It has been successfully applied to both discrete and continuous features of high-dimensional data. OSFMI is primarily compared with the rough set, as the latter requires quadratic computational time $O(n^2m)$ and can only be applied using discrete features [137]. The author also claimed that OSFSMI could effectively distinguish the most informative and non-redundant features that were relevant to the target class and that it ranked among the best in terms of accuracy, running time, and stability.

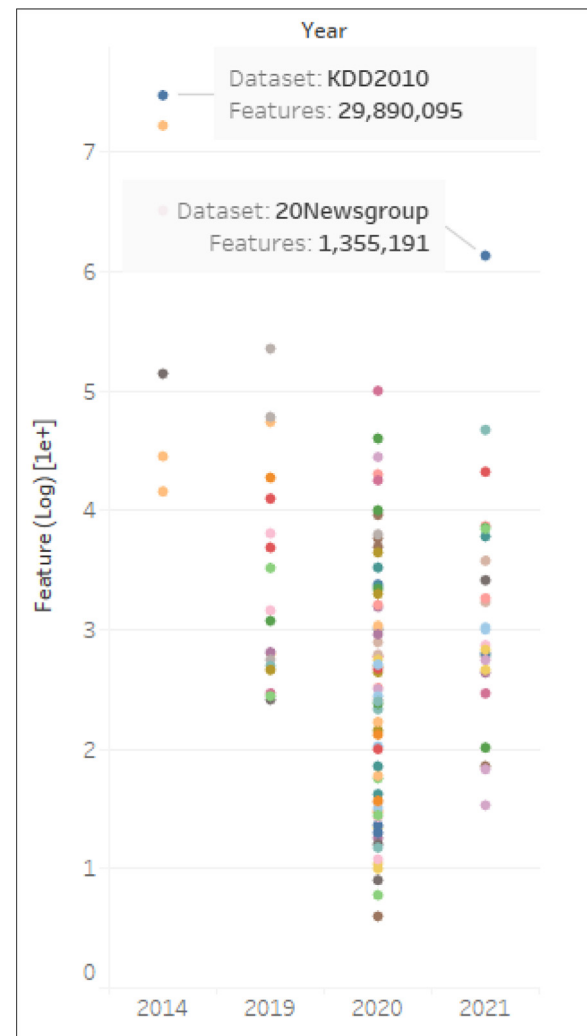


Fig. 7. The dimensionality of features experimented on using OFS methods (over three years).

OSMWFS was incorporated with the fuzzy rank aggregation method to determine the final ranking of features in the candidate feature sets. The fuzzy fast-minimum redundancy and maximum relevance feature selection method (Fast-mRMR) was then further utilised on multiple windows to improve the performance. Fast-mRMR used MI to select strongly relevant and non-redundant features. OSMWFS outperformed other methods when handling high-dimensional data using parallel computing and sliding windows [159].

Fahad et al. [135] employed symmetric uncertainty (SU) on ACO-SU, a modified version of information gain (IG), for the heuristic search of ACO on a large set of features of medical images. A correlation-based filter method, SU is more efficient than IG in eliminating redundant features because pairs of features are handled symmetrically. Since information gain is more inclined towards features having a greater number of values, symmetric uncertainty measures the exclusivity of each feature in the proposed approach. ACO-SU requires features with SU above the defined threshold to assign the weight and initial population.

MOANOFs maximises the prediction error and time cost while maximising the agent learner trust when using multi-objective agent adaptation for an ultra-high-dimensional database. MOANOFs utilises the Automated Negotiation (AN) technique by allowing Intelligent Agent-Learners to allocate weight values

to relevant features and 0 values to redundant features [42]. However, it is still unclear how the negotiation of redundant feature values is achieved. The MOANOFs system has been proven to achieve high accuracy with several large-scale real-text classification datasets, such as 20Newsgroups and RCV1.

The Levy Flight Bat Algorithm (LFBA) utilises OFS to choose significant features to overcome the Frequent Pattern Mining (FPM) computational challenge and improve the classification accuracy of sparse and high-dimensional datasets [160]. The LFBA adopts the Bat Algorithm (BA) as the optimisation method to indicate the optimal selected feature solution, based on bat locations that are indicated through echolocation.

Renuka and Sasikala [133] proposed the OFS-ABA to enhance classifier efficacy for applications that stream features in advance and have indefinite knowledge of feature space. The OFS-ABA employs the Accelerated Bat Algorithm to find the optimal streaming features, based on the echolocation activities in bats' behaviour. The OFS-ABA is designed to handle the volume, variety, and velocity of data and employs MapReduce as the processing method.

BS-SFS improves the Grafting approach by exploiting the Bayesian penalised method to estimate the regularisation parameter adaptively when a new feature arrives [118]. As a result, even if the feature stream size is unknown, BS-SFS can be applied to a broader range of predictive models and is not limited only to classification problems. Despite selecting more features than some benchmark OFS methods, BS-SFS handles high-dimension micro-array datasets effectively by selecting a small number of features.

4.4.3. Online feature selection on multi-label

In classical supervised learning, each object belongs to a candidate class. Unlike traditional single-label learning, some objects in multi-label learning are associated with multiple concepts simultaneously, which leads to difficulties in anticipating all the relevant labels for a given object [114]. To exacerbate this, the issues of high dimensionality and streaming data must be addressed by multi-label learning due to the unknown feature space with highly irrelevant and redundant features [85,162,169]. Multi-label feature selection with streaming labels assumes that the labels are no longer static but flow in one by one over time. Hence, feature selection must be executed when a new label arrives [113]. Different search strategies have been determined to be able to handle multi-label learning, some of which handle group stream data in similar ways.

A multi-label streaming feature selection approach based on a neighbourhood rough set, OM-NRS, has been proposed that granulates all instances using maximum-nearest-neighbour and employs a generalised neighbourhood rough set in single-label to fit multi-label learning [161]. The drawback of OM-NRS is that it requires the parameter to be specified as the influence over the relevance threshold and threshold settings varies for different datasets. Lv et al. proposed OFSI to consider feature interaction based on a neighbourhood rough set [123]. A rough approximation model based on maximum neighbourhood was designed to cater for the downside of traditional feature selection methods in handling feature interaction on multi-label data. Online significance and online redundancy criteria were designed by retaining highly interactive features and discarding redundant ones. However, the experiment demonstrated that OFSI was inefficient in handling some real-world datasets, such as drug repositioning and image analysis, which involved features arriving by the group.

MI has been employed in FSSL to obtain the statistical correlation between two random variables handled on streaming multi-label learning [113]. By introducing inter-class discrimination and intraclass neighbour recognition, FSSL demonstrated a

competitive efficiency and effectiveness performance compared to other state-of-the-art multi-label feature selection methods. Lin et al. [162] proposed a multi-label online streaming feature selection algorithm based on fuzzy mutual information called Multi-label Streaming Feature Selection (MSFS). The authors indicated that MI can analyse correlations between feature subsets and prediction targets. On the contrary, it fails to consider the redundancy in feature subsets. Hence, fuzzy mutual information was proposed to handle this drawback.

It was proposed that OMGFS [41] be used in two phases to handle group and multi-label datasets: online group selection and online inter-group selection. In the first phase, OMGFS used neighbourhood mutual information to measure the correlation between two features, after which symmetrical uncertainty was used to measure the relevancy between features with multi-label. Since this phase did not consider information between groups and neglected redundancy and interaction, the second OMGFS phase was constructed to handle redundancy and interaction with a different group. By defining a theorem in redundancy analysis and using the interaction weight, OMGFS was able to filter redundant features during the inter-group selection and return optimal feature subsets through weight updating. Hence, OMGFS demonstrated the ability to handle feature selection dynamically, compared to other benchmark multi-label feature selection methods with a better stability index. However, the performance of OMGFS was influenced by the relevance threshold defined by the number of selected features.

I-SFS and G-SFS were proposed to handle online feature selection when the features arrived individually and in group form on multi-label datasets. By adopting the multi-objective cuckoo search method, multiple objectives were considered in this research: (1) the relevancy of selected features with respect to the class label and (2) the redundancy of selected features with respect to other features [136]. When compared to the single-objective optimisation, it was claimed that I-SFS and G-SFS performed better as multiple properties were considered in the feature selection process. I-SFS and G-SFS adopted the cuckoo search because fewer parameters need to be tuned than when using other benchmark optimisation techniques like Particle Swarm Optimisation and the Genetic Algorithm.

In SF-DSW-FRL [158], neighbourhood conditional mutual information was used, and minimum neighbourhood conditional mutual information was later incorporated. SF-DSW-FRL was able to measure the redundancy between the relevant candidate features and the selected feature subsets for multi-label data. To cater for group feature processing, the author used hybrid fixed and dynamic sliding windows. SF-DSW-FRL required an adjustment of the dynamic window threshold for optimal features to be saved into the selected feature subset. Pareto-FS is a fast, accurate multi-label online feature selection method that can handle sequentially added features [163]. Pareto-FS uses symmetrical uncertainty (SU) to find the relationship between features and labels. The SU concept had previously been defined by the information gain and entropy method to find the complete dependency of features and classes. Features with lower SU values were considered for the redundancy and irrelevancy analysis. Pareto-FS handles multi-label feature selection through a multi-objective optimisation approach, in which correlations between features and labels are considered objective functions. Recognised as a method that does not require an optimal threshold to select the best features, Pareto-FS has demonstrated accuracy and outperformed some benchmark multi-label feature selection methods by classifying within an appropriate running time.

4.4.4. Online feature selection on class imbalance

In the case of class imbalance, [164] proposed K-OFSD, having been motivated by the challenge of high-dimensional and class-imbalanced data. The author mentioned that in high-dimensional data, the class distribution skewness tends to be overwhelmed by the large classes, resulting in high accuracy since all the instances are classified into particular large classes. Meanwhile, in the medical field, people require more analysis of abnormal cases than common cases. Hence, by leveraging the nearest neighbour information, K-OFSD selects relevant features that can improve the separation between the majority and minority classes. According to maximal dependency, maximal relevancy, and maximal significance evaluation criteria, OFSD demonstrated a selection of features with high correlation, high dependency, and low redundancy. According to the author, although the dependency evaluation requires a prior threshold definition, the running time can be reduced.

The OSFS-KW proposal involved the exploitation of two main types of neighbourhood relations: (1) the neighbourhood relation with fixed neighbour (k -nearest neighbourhood relationship) and (2) the neighbourhood relation with fixed distance (δ neighbourhood relation). This considered the structure of the neighbourhood covering without the need for parameter adjustment [131]. By proposing a weighted dependency degree that considered the structure of neighbourhood covering, it was claimed that OSFS-KW could overcome the challenge of using OFS with medical data, which contains an uneven distribution of datasets and class imbalance.

Meanwhile, MOFS was proposed to tackle the problem of large-scale imbalanced datasets in online feature selection by adapting the cost-sensitive learning approach [165]. Motivated by the challenge that a single-label dataset is always imbalanced and affects the classification result, in MOFS, Binary Feature Selection was adopted for imbalanced datasets using the basic principle of cost-sensitive learning, thus imposing greater weight on features that are significant for predicting positive examples. Regularisation parameters are also defined to control the complexity of the model.

4.4.5. Online feature selection on capricious data

The assumption has been made in recent OFS studies that a fixed feature space is involved. Unfortunately, in real applications, streaming data features vary arbitrarily, which has been precisely defined as capricious data streams. Incoming instances may not carry some features that had been included in the global feature space, known as random missing features [170]. Latent Factor Analysis and Markov Blankets have been used to handle capricious data. Meanwhile, referring to Table 6, GF-CSF aims to handle a single flow of capricious streaming features using latent factor analysis (LFA) in the pre-processing phase to complete the missing entries [166]. Based on this features completion, GF-CSF then uses irrelevant feature definitions for the online relevancy analysis. Meanwhile, for the online redundancy analysis, the author adapted the definitions of the Markov Blanket and the redundant complete streaming feature to remove redundant features. However, GF-CSF requires extra computational time compared to other benchmark methods to complete the missing values of capricious streaming features.

4.4.6. Online feature selection on causal-based

In many research and technology fields such as health, biology, and pharmacology, causal discovery has a practical significance. The purpose of causal discovery is to uncover causal relationships between features. One significant example of causal discovery is determining the cause of disease to facilitate treatment and prevention [171]. In streaming data, when the full feature space

is unavailable before learning begins, uncovering the causal relationships between features is a challenge when dealing with this form of data behaviour. Hence, the following OFS methods have been proposed; the review indicated that most of the methods applied are hybrids.

OSFSW was designed by You et al. as a solution to overflow streaming features with high redundancy and a causal relationship [147]. By using Markov Blankets (MB) and Sliding-windowing, this research claimed to have employed fewer selected features with higher accuracy than SAOLA and Alpha-Investing. The Markov Blanket approach was adopted to retain strongly relevant features and remove those that were irrelevant and redundant from the feature space. Sliding-windowing was imposed as a buffer to gather relevant features continuously before the features were handled in the redundancy analysis. You et al. then proposed another method, named OMBSF, motivated by the challenge that the traditional Markov Blanket methods were unsuitable for streaming features [52]. OMBSF was proposed to address the causal-based dynamic feature space in the case of high-dimensional data by using online MB discovery to eliminate irrelevant and redundant features. A Bayesian network was embedded to trace the conditional relevancy and independency of the features that arrived, based on conditional independencies. The authors declared that OMBSF performed better in handling causal-based features in highly synthetic and real-world datasets.

SDMB was proposed to handle the challenge of summarising historical with causal learning streaming data for incremental MB learning in one scan. Since new instances are constantly being added in streaming data behaviour, the distribution may shift over time and result in a feature that is a member of the MB of the target features not being so at other times. Having employed a dynamic AD-tree to sequentially learn the MB of the target when new data arrives, the author reported that SDMB outperformed other state-of-the-art methods, especially on a real-world dataset. Nevertheless, SDMB was unable to handle high-dimensional streaming data due to its methodology of reconsidering discarded features every time new features arrive [87].

CSSU was designed as a stream feature-based causal learning algorithm that obtains candidate neighbours with symmetrical uncertainty (SU) to avoid subset searches before executing the conditional independence (CI) test [167]. By adopting SU and CI, CSSU significantly reduces the time complexity and improves the accuracy. CSSU heuristically selects potential candidate neighbours via the optimal threshold obtained through a series of experiments. CD-SF was proposed and optimised as CD-SU-SF, which incorporated Symmetrical Uncertainty into MI to calculate real-dependent relationships in causal-based data using a dataset of lung images. By employing feature evaluation instead of subset evaluation, the number of conditional independence calculations of CD-SU-SF was greatly reduced. The drawback here is that MI thresholds need to be determined by the users to obtain the candidate neighbour nodes [132].

4.4.7. Online feature selection on low redundancy and high relevance

The benchmark OFS methods such as Alpha-Investing, OSFS, and SAOLA suffer from inadequate prediction accuracy or running time if the streaming features have low redundancy and high relevance characteristics, such as in real-time medical diagnosis [149]. To address this behaviour, several methods have been proposed that use the Markov Blanket and hybrid techniques.

You et al. proposed an improved OFS method to handle the effects of low redundancy and high relevancy data behaviour by developing ConInd [149]. This research also undertook relevancy and redundancy analysis using a Markov Blanket algorithm. The

Markov Blanket was optimised by adapting three-layer filtering comprised of (1) null-conditional independence for filtering irrelevant features, (2) single-conditional independence to discard parts of the redundant features in weakly relevant features, and (3) multi-conditional independence to remove the remaining redundant features. Based on the authors' experiment, ConInd was able to reduce the running time to 53.56 per cent compared to when OSFS was used (a benchmark OFS method). However, if more relevant features are introduced, ConInd experiences uncertainty in the feature space structure and requires fine-tuning to avoid learning rate decay.

OSFSCM has been proposed to optimise the selection of low redundancy and high relevancy feature subsets by exploring the OSFS method [130]. In the first phase of OSFSCM, irrelevant features were filtered using non-conditional independence. MI theory was then applied in the second phase through Fisher's Z and G^2 tests to find the correlation between features and filter the partially redundant features in continuous and discrete data, respectively. Subsequently, in the last phase, conditional independence was adopted to remove the remaining redundant features. OSFSCM was shown to select more features, based on its impressive performance in mining the elements from Markov Blankets while obtaining high accuracy and efficacy, compared to Alpha-Investing and OSFS. When dealing with causal-based datasets with low redundancy and high relevancy (strong and weak relevancy), OSFSCM proved to run faster than OSFS.

4.4.8. Online feature selection on feature drift

When dealing with streaming data, the terms incremental learning and data stream learning were coined together. Feature selection refers to identifying the features relevant to a learning task [57]. However, in real stream data behaviour, not only do feature changes occur but also new classes can occur from the stream without prior knowledge, which suggests the need for a redefinition of the used model [172]. Data streams require incremental learning, in which the learner incrementally identifies which features are relevant to the label while the data changes over time. This situation is defined as feature drift. Feature drift has received little attention from researchers, and a new method must be designed to decide which features are the most relevant, subsequently taking advantage of the learning process [46,173]. Feature drift, which developed from the theory of concept drift, impairs the learning process of streaming data since considerable variations are involved in the data distribution.

OFSEER was presented to tackle the presence of feature drift in online feature selection. Using evolving regularisation on the existing OFS algorithm, OFSEER dynamically penalised model complexity, thus reducing the impact of feature drift in the feature selection process. An OFSEER experiment was conducted on the text stream feature space due to the changes of relevant terms that were subject to time and the real-time processing of the classifier was required to maintain an accurate model [61,174]. The author also mentioned that the requirement to balance the regularisation rate depended on the irrelevant and redundant terms available.

4.4.9. Other data behaviours

Other data behaviours also require greater research attention, such as noisy data and structure data. It is evident that less research has been conducted on these behaviours due to the challenges. A classical rough set-based online feature selection named OS-NRRSAR-SA was proposed by Eskandari and Javidi [129]. This set requires no parameter specification and uses a noise-resistant dependency measure. Nevertheless, it cannot handle numerical data. SFS-RS implements the classical rough set-based feature significance concept to enhance the elimination of irrelevant features. Then, OSFS-MRMS was designed by

adapting the OSFS framework to optimise OS-NRRSAR-SA [150]. OSFS-MRMS enhanced the redundancy analysis method, based on the rough functional dependency concept.

4.5. Evaluation metrics: Impact of threshold initialisation, dimensionality, and relevancy and redundancy measures

The performance of OFS methods have been frequently evaluated using k-NN, SVM, and Decision Trees to evaluate the accuracy, running time, and number of selected features for single-label OFS [130,150]. Meanwhile, different evaluation metrics have been used in Multi-label OFS: average precision, ranking loss, coverage, One-Error, Hamming Loss, and Micro-F1 [162,175]. The Friedman [176] and Bonferroni-Dunn tests [177] are usually adopted to further explore the statistical significance among the experimented OFS methods. Using the review of the state-of-the-art methods in the previous section, the authors further characterised and analysed the best search strategy methods for handling single-label OFS by examining different performance measures, as depicted in Table 7. These performance measures, accuracy and running time, were undertaken by examining the impact of threshold initialisation, dimensionality, as well as the levels of relevancy and redundancy of the dataset. The OFS methods excluded multi-label data because accuracy was the aim. Meanwhile, multi-label OFS utilises different evaluation criteria, as presented in Table 7.

4.5.1. Threshold initialisation vs accuracy vs running time analysis of OFS methods

As demonstrated from the review of the state-of-the-art methods in the previous section, the current authors characterised and analysed the best search strategy methods that can handle OFS with satisfactory accuracy, as demonstrated in Table 7. Of all the methods outlined and discussed, the Markov Blanket, Neighbourhood Rough Set, and Information Theory search strategies were chosen for inclusion in the performance analysis of these OFS methods. This was because these techniques demonstrated the highest average accuracy. Based on the analysis presented in Table 7, the Markov Blanket algorithm showed the highest average accuracy of all the algorithms that adopted this technique, with a value of 0.86, followed by the Neighbourhood Rough Set and Information Theory-Based techniques, with average accuracy levels of 0.8 and 0.76, respectively.

The evolutionary algorithm (EA), multiobjective, windowing, and regularisation techniques are some of those that have been introduced to be hybridised with the available search strategies to improve the streaming data handling performance. EA and windowing show the highest accuracy of all the techniques. EA was applied to acquire the best feature subset and concurrently to increase the performance measure [179]. When EA was applied by past researchers, the average accuracy rose to more than 0.94, with OFS-ABA [133] achieving 0.98 accuracy and LFBA [160] achieving 0.91.

Meanwhile, windowing was hybridised to handle high dimensionality. OSFSW [147], an OFS method that hybridises Markov Blankets and windowing, handled high redundancy with causality and produced a significant rise in accuracy to 0.95. HOSFS [63] combined windowing with the heuristics function, resulting in 0.90 accuracy. However, due to the lack of research applying EA and windowing to OFS methods, the efficiency of these techniques could not be determined, unlike other baseline search strategies that have received more attention.

Since the analysis accuracy was insufficient to conclude the performance measure of the OFS methods, they were then further

Table 7

Characteristics and performance of OFS methods (Single-Label OFS).

Search strategy	Purpose	Method	Requires domain knowledge	Require threshold initialisation	Accuracy	Average running time	Highest features (Dimensionality)	Running time	Accuracy of highest features	Dataset
Information theory -based & Windowing	Feature relevancy & redundancy	OSMWFS [159]	No	No	0.95	41.92	47236	100	0.90	rcv1
Information theory-based	Feature relevancy & redundancy	OSFSMI [137]	No	No	0.53	26.88	49151	65.65	0.60	GLA-BRA-180
	Feature relevancy, redundancy, interaction	SFS-FI [178]	No	Yes	0.86	3.95	24481	3	0.50	breast
	Feature relevancy & redundancy	OGSFS-FI [55]	No	Yes	0.8	22.07	24481	3.85	0.60	breast
	Feature relevancy, redundancy	CD-SF & CD-SU-SF [132]	No	Yes	0.7	138.18	1017	138.18	0.70	Hailfinder
	Feature relevancy, redundancy	OSFSCM [130]	No	Yes	0.77	22.12	19993	12.16	0.70	SMK_CAN_187
	Average accuracy				0.76					
Markov blanket	Feature relevancy and redundancy	SDMB [87]	No	No	0.86	2	42	19.28	0.69	Connect4
	Feature relevancy and redundancy	ConInd [149]	No	Yes	0.89	99.5	19993	327.15	0.89	SMK
Markov blanket & Windowing	Feature relevancy and redundancy	OSFSW [147]	No	Yes	0.95	35.01	19993	22.33	0.69	SMK
Markov blanket	Feature relevancy and redundancy	OMBSF [52]	No	Yes	0.87	77.67	20000	180.12	0.89	Dexter
	Average accuracy				0.89					
Neighbourhood Rough Set	Feature relevancy	OSFS-KW [131]	No	No	0.88	66.29	17816	114.49	0.70	Breast Cancer
	Feature relevancy & redundancy	OFS-A3M [153]	No	No	0.84	2.7	12533	29.43	0.90	Lung
	Feature relevancy & redundancy	OFS-Density [124]	No	Yes	0.8	5.7	10000	36.23	0.80	Arcene
	Average accuracy				0.8					
Evolutionary algorithm	Feature relevancy	OFS-ABA & EIDMLP [133]	No	Yes	0.98	0.74	100000	0.07	0.98	Dorthea
	Feature relevancy	LFBA [160]	No	Yes	0.91	448	36	639	0.93	Chess
	Average accuracy				0.94					

characterised according to threshold and non-threshold parameter initialisation. OFS methods that require threshold initialisation, λ , are defined as threshold-based methods. For high-dimensional real-world datasets, the interaction gain value could be very small for many new arriving features. Thus, a threshold had to be imposed to reduce the number of selected features, reduce the time, and improve the efficiency. However, when the λ values increased, the corresponding running time increased rapidly because larger λ values needed more time to run the feature redundancy analysis.

An attempt was made to map the relationship between threshold-based method, accuracy, and runtime, as detailed in Fig. 8. For the eight threshold-based OFS methods (SFS-SI, OGSFS, CD-SF, OSFSCM, ConInd, OSFSW, OMBSF, and OFS-Density) shown in this figure, the average accuracy was 0.839. Meanwhile, for the five non-threshold-based methods (OSMWFS, OSFSMI, SDBM, OSFS-KW and OFS-A3M), the average accuracy was slightly lower at 0.812. These analyses indicate that imposing a threshold does not significantly affect the accuracy of OFS methods. However, threshold initialisation generally reduces the runtime but not for all the OFS methods. OFS-A3M [153] could still perform with a low runtime and with reasonable accuracy without imposing a threshold, whereas OSMWFS [159] resulted in high accuracy, but the trade-off was a higher running time. Based on Fig. 8, it could not be concluded that threshold initialisation directly impacts the runtime since further analysis was required from different perspectives: dimensionality and the relevancy and redundancy levels of the datasets.

4.5.2. Analysis of Running Time and Dimensionality using OFS methods

The next step was to analyse the highest features experimented on, using every OFS method and the running times to ascertain the impact of dimensionality by examining various obvious results. As seen in Fig. 8, SDBM showed a very low running time, 2 s, with a reasonably acceptable accuracy value of 0.86. Nevertheless, after further investigation, as illustrated in Fig. 9, the dimensionality of SDBM confirmed that its low running time was due to its low dimension, with a maximum of 42 features being used, whereas other methods used thousands of features.

An attempt was made to compare OSFSW, ConInd, and OSFSCM since these methods experimented on the SMK dataset (19,993 features) as the highest dimension dataset. ConInd showed a very high running time of 327.15 s, while OSFSCM achieved only 12.16 s. The author verified that with the increase in the feature size, ConInd was unsuitable for dealing with high redundancy and low relevance datasets [149]. In general, dimensionality increased the running time. However, this is not a direct relationship as it actually depends on how the algorithm is developed, related to the computational time for every segment of the algorithm, as well as the level of relevancy and redundancy of the data. Overall, OSMWF (non-threshold-based) and OSFSW (threshold-based) are two outstanding methods that can find a compromise between accuracy and running time, with both methods having an accuracy of 0.95.

Therefore, based on the analysis, a general relationship was deduced, as shown in Fig. 10. The runtime was affected by three different measures: dimensionality, threshold, and relevancy and redundancy. When dealing with higher dimensionality, a longer runtime will be the trade-off. Generally, when a threshold was imposed, the runtime decreased. However, larger λ values did not mean better performance [149,180]. The threshold value also impacted the runtime; the higher the threshold, the longer the runtime. Under certain circumstances, for example, when dealing with redundancy analysis, larger λ values mean more time is needed to run the feature redundancy analysis. OFS methods

Table 8

Tools used in OFS methods based on data behaviour.

Data behaviour	Tools			
	LOFS	MOA	LIBOL	KEEL
Single-Stream	/	/		
Group Stream	/			
Class Imbalance	/			
Sparse Data	/			
Causal-Based	/			
HDD	/	/	/	
Ultra-high-dimensional	/		/	
Feature Drift		/		
Multi-label				/

unable to handle high redundancy and low relevance datasets effectively experienced longer running times [149,164]. In fact, the justification was based on how the algorithm of each method is developed, considering the computational time for each segment for which the method aims, for instance, relevancy analysis and redundancy analysis, as well as data behaviour.

4.6. Tools

Also explored were several open-source toolboxes for online feature selection that were applicable for both research and practical applications. Originally, the framework for each tool was designed to aid collaboration between research groups; these subsequently allowed researchers to test their theories. Based on the review of the published articles, different tools can handle different data behaviours on streaming data, as depicted in Table 8.

4.6.1. LOFS

Designed and developed by Yu et al. [181], LOFS was the first comprehensive open-source library created for Matlab and OCTAVE to implement state-of-the-art online streaming feature selection techniques. LOFS tools encourage academics to enhance LOFS and share their algorithms using the LOFS framework, facilitating the development of new novel online algorithms for handling the considerable challenge of high dimensionality in big data analytics. The LOFS architecture is built on three main modules: (1) the Correlation Measure (CM), (2) the Statistical Correlation Measure (SC), and (3) the learning module. Two learning sub-models are included in the learning module, LFI for single-stream data behaviour consisting of Alpha-Investing, OSFS, Fast-OSFS, and SAOLA; and LFG for group stream behaviour, comprising group-SAOLA. According to past publications, LOFS is used by researchers to test not only single and group streams [145] but also class imbalance [71,131], sparse data [124,145], causal data [157], and high-dimensional data [153].

4.6.2. MOA

The MOA framework provides various algorithms for multiple tasks related to data stream analysis [182]. This framework allows the import of an existing dataset and its simulation as a continuous flow of data. The framework is connected to a large community of researchers who regularly add new algorithms and challenges to it. Classification, regression, multi-label, multi-target, clustering, outlier identification, concept drift detection, active learning, and other tasks are featured in MOA. MOA includes three modules: (1) data generators (such as AGRAWAL, Random Tree Generator, and SEA), (2) evaluation methods (such as periodic holdout, test-then-train, and prequential), and (3) statistics (CPU time, RAM-hours, and Kappa). MOA has a GUI (Graphical User Interface) and a command line interface, facilitating batches of tests. The implementation is written in Java

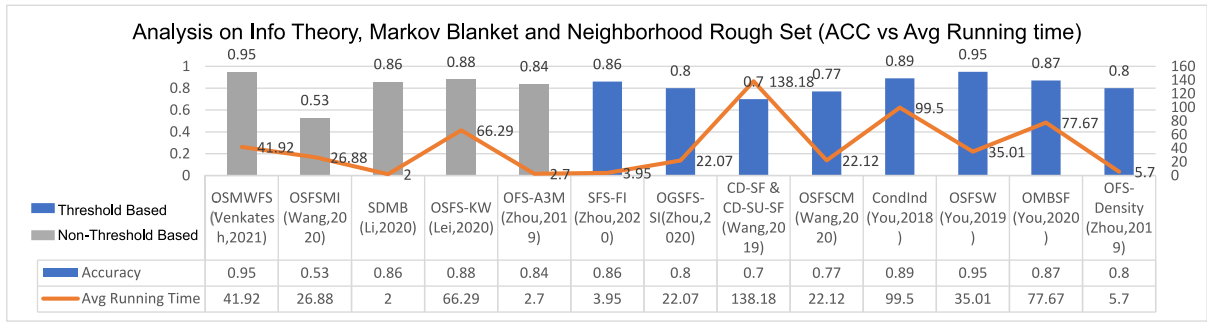


Fig. 8. Analysis of OFS methods – Accuracy vs Average running times.

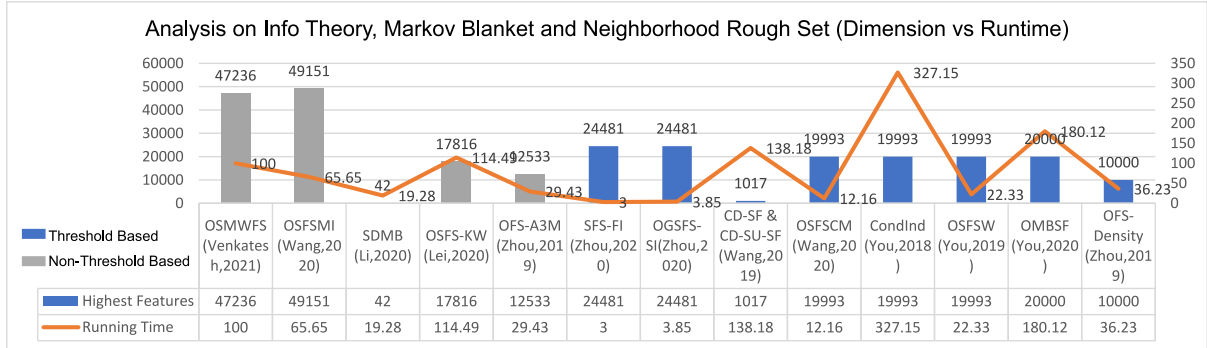


Fig. 9. Analysis of OFS methods – Highest feature vs Running times.

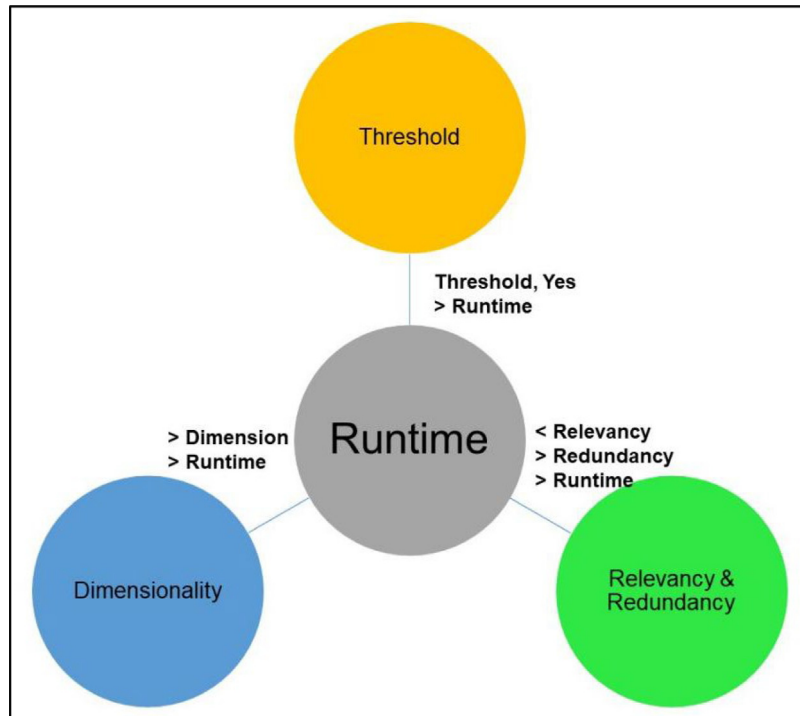


Fig. 10. Relationship between dimensionality, threshold, relevancy and redundancy, and running time.

and it shares many features with the WEKA framework, including the ability to extend the framework by inheriting abstract classes. MOA is employed in state-of-the-art studies, including OFSER [61] and HOSFS [63], which investigate single-stream, high-dimensional data and highlight drift data behaviour.

Meanwhile, Massive Online Analysis Feature Selection (MOAFS) [174] is a MOA package that performs feature selection

in classification situations involving data streams. MOAFS was developed to perform various experiments on some of the most relevant feature selection strategies presented for data stream classification challenges. MOA contains seven feature selection algorithms that are used as dimensionality reduction techniques in data stream classification problems, especially in the text-domain field, since they are not directly available on MOA. The feature

selection methods in MOAFS include information-based filter algorithms (Info Gain and Gain Ratio), symmetrical uncertainty, chi-squared, OFS and Extremal Feature Selection (EFS) [61].

4.6.3. LIBOL

LIBOL is used to run OFS experiments dealing with high-dimensional and ultra-high-dimensional data with large RAM requirements [42,183]. LIBOL is a large-scale online learning open-source library that includes a broad family of efficient and scalable state-of-the-art online learning algorithms for large-scale online classification applications. The aim of LIBOL is to handle the big data analytics challenge, dealing particularly with machine learning in massive data streams [184]. LIBOL has been used in studies [42] to investigate high-dimensional data and [183] experiment on ultra-high-dimensional data.

4.6.4. KEEL

KEEL (Knowledge Extraction based on Evolutionary Algorithms) is a JAVA software tool that implements a wide range of feature selection methods, such as Mutual Information, ReliefF, ensemble methods and evolutionary algorithms [185]. KEEL can assess evolutionary algorithms for various data mining issues like regression, classification, and unsupervised learning. KEEL 3.0 was then released with four new learning modules, and it also includes a dataset repository, containing specifically multi-label, imbalanced, noisy, time series, and semi-supervised data [186]. A non-parametric statistical analysis module is also included in KEEL, which is useful for evaluating the statistical tests for pairwise comparisons in feature selection research [25,29]. Kashef et al. used KEEL to review and experiment on multi-label feature selection [64] and extend the research by [163] to evaluate evolutionary algorithms applied on OFS and handling multi-label data, resulting in superior classification performance and a faster algorithm.

4.6.5. Tool performance analysis on HDD and Ultra-HDD

According to Table 8, different tools can be applied with different data behaviours. Tools like LOFS and MOA can manage various data behaviours, while others, such as LIBOL, can only handle specific data behaviours like high-dimensional and ultra-high-dimensional data. KEEL is used specifically for multi-label data in OFS.

LOFS, MOA, and LIBOL are capable of handling HDD data behaviours. Hence, the performance analysis of all three tools was compared based on the results of state-of-the-art studies. The analysis was also extended to ultra-high-dimensional (Ultra-HDD) behaviour that can be handled by the LOFS and LIBOL tools. Table 9 illustrates that LIBOL successfully achieved its aim of handling massive data streams because it was the most robust tool for handling HDD and Ultra-HDD, attaining the highest accuracy and runtime results. Other authors [42] have experimented on HDD using LIBOL and achieved more than 95 per cent accuracy. In addition, in [183], the author employed the SOFS method with the LIBOL tool on Ultra-HDD and obtained up to 90 per cent accuracy and lower runtimes that ranged below six seconds. This can be compared to the results when the same dataset (News20) was used by [145], who employed four benchmark OFS techniques with LOFS tools — Alpha-Investing, Fast-OSFS, OFS, and SAOLA — and attained an average accuracy of 73.2 per cent and a runtime of 1326 s. Using the MOAFS technique, the MOA tool achieved an accuracy range of 79.53 per cent to 85.71 per cent and a high runtime of 343 s or above on an HDD dataset [61]. Even so, the datasets used, SpamAS and Usenet3, exhibited feature drift behaviour in addition to HDD behaviour. In comparison, the LIBOL tool outperformed others in relation to HDD and Ultra-HDD data. However, it should be noted that the design of the OFS techniques also impacted the results, as depicted in Table 9.

5. Discussion

This section contextualises the challenges derived from the reviews and the proposed framework, as shown in Fig. 5. This framework offers added value to this study by supporting further exploration in the OFS field.

The first challenge determined in this study was data behaviour. Of all the data behaviours studied, more attention should be directed to the real-world data behaviours of streaming data like concept drift, multi-label, and imbalanced data. Concept drift is defined as a concept or topic that appears and disappears frequently. The possibility of it reoccurring or otherwise is uncertain. Concept drift represents a challenge in the feature selection process as feature drift changes with time. Feature drift occurs particularly whenever a subset of features becomes relevant to the concept that is to be learned in the predictive model [187]. A highly prominent research domain of feature drift is text mining, for example, handling spam email detection based on the unavailability of feature space and the submission of tweets in real time as an event happens [61]. Although the current techniques for data stream classification handle most of the challenges posed by streaming environments, little attention has been given to feature drift. As the framework outlines, OFSER was one of the proposed OFS methods to deal with feature drift by using the regularisation method as a correlation measure to regularise the relevancy subject to time, while MOA would be used as the OFS tool to deal with the data stream and drift. However, this method requires real-time processing on the classifier to maintain an accurate model. This was supported by Barddal et al. [188], who suggested that to maintain an accurate predictive model of a data stream that exhibits feature drifts, a classifier must be trained and updated on the set of features that is currently relevant. Zhao and Koh [187] mentioned that further investigation was needed to develop an interpretation system for feature drift. Hence, future research should focus on detecting, adapting, and interpreting changes in feature relevancy for feature drift.

Next, multi-labels have been identified as another OFS issue and an aspect of the data behaviour challenge. Multi-labels exist in multiple domains and real-world applications, such as social media networks, spam email filtering [189], text categorisation [65], image analysis [190,191], and gene function classification [41]. Multi-label learning requires learning with multiple concepts and faces the challenge of distinguishing relevant, irrelevant, and redundant features in data streaming. Multi-labels have become an interesting area of study because they involve real-world problems and are subject to many instances. As shown in Table 6, solutions like OM-NRS [161], MSFS [162], and OMGFS [41] employ neighbourhood rough sets, information theory, and symmetrical uncertainty as correlation measures and further improve the feature relevancy, redundancy, and interaction by selecting the relevance threshold. Furthermore, optimisation methods [136,163] have been proposed to enhance relevancy and redundancy in feature selection. However, time complexities and group streaming have been identified as further challenges in handling multi-label data and which relate closely to feature drift behaviour. Hence, in future, the studies on relevancy, redundancy, and irrelevancy should be expanded by combining hybrid methods of optimisation in selecting the relevance threshold.

Like multi-label learning, class imbalance is also a common data behaviour of OFS that has received little attention [192]. Medical diagnosis [193] and fraud detection [144,194] are among the real-world domains that exhibit high-dimensional data with high class imbalance since more instances of classes exist compared to others. Solutions like K-OFSD [71] and OSFS-KW [131], which were discussed in Section 4.4.4, used neighbourhood rough sets to handle imbalanced classes. However, both methods faced

Table 9

Tool performance measure based on data behaviour.

Data behaviour	Paper	Dataset information		Performance measure	
		Dataset	Dimensions	Accuracy (%)	Runtime (s)
Method: LOFS					
HDD	[145]	Dexter	20,000	64.5	3.5
		Dorothea	100,000	87.8	229
Ultra-HDD		News20	1,355,191	73.2	1326
		Webspam	16,609,143	95.25	10,499.5
Method: MOA					
HDD	[61]	SpamAS	40,000	85.71	586.709
		Usenet3	27,893	79.53	343.507
Method: LIBOL					
HDD	[42]	Pcmac	7,510	> 95	71.4
		basehock	4,863	> 95	69.2
Ultra-HDD	[183]	News20	1,355,191	91.26	0.61
		rcv	47,152	90.40	6.29
		URL	3,231,961	98.32	6.95

runtime issues, with K-OFSD requiring a prior threshold definition to reduce the runtime and OSFS-KW demonstrating a slower runtime compared to the other methods experimented on. Nevertheless, imbalanced data always affects the evaluation metrics if handled improperly. Of all the methods studied, only MOFS [165] tackled the evaluation measure issue by counting the number of mistakes made in the learning process using specificity and the geometric mean. Hence, these solutions could be further investigated to address high class imbalance with better evaluation measures.

The next challenge in OFS concerns data quality. Two data quality issues affecting real-world applications are missing attributes and noisy data. Section 4.4.9 referenced the resolution of noisy data. OS-NRRSAR-SA [129] used a classical rough set of correlation measures on a 24 GB computer memory to enhance the performance of the algorithm. Consequently, Javidi optimised [150] the algorithm of OS-NRRSAR-SA, developing OSFS-MRMS to improve the redundancy analysis by using information theory with classical rough sets. In handling missing data, the recent OFS-Mean solution [67] used generalisation ability with neighbourhood rough sets for no prior knowledge of feature space. All these proposed methods used rough set theory to address data quality issues. Nonetheless, the evaluation measures of missing and noisy data remain an interesting topic for future research.

Subsequently, when streaming data is continuously and rapidly changing, it affects the unsupervised method in the underlying structure of a cluster that continuously varies and alters. Solutions like SFGS [53], discussed previously in Section 4.4.1, employ clustering-based feature selection to handle group feature selection. This is done by defining clusters based on the creation of the grouping. This method performs a user-defined parameter to control the distance threshold when building a new cluster or adding a new feature to the existing one, but it requires a measure to perform on the real behaviour of constantly changing streaming data. Next, when handling instances arrival for unsupervised methods, the FSMCP [121] solution has tackled unstable associations of data structures by minimising the differences between the multi-cluster structures in the original and the selected feature space. However, future unsupervised OFS studies must be conducted on the efficiency of drift detection methods for streaming data, in which labels arrive with a non-negligible delay or when some or all labels never arrive in either semi-supervised and unsupervised learning [57]. Besides, it is costly to apply this unlabelled data manually [195]. Hence, the area of unsupervised OFS remains an area yet to be extensively explored and studied.

Finally, high-dimensional data (HDD) is the most problematic challenge defined in most OFS research [22,40,51]. This article considers four main issues of HDD in OFS: scalability, stability, computational cost, and distributed processing in a dynamic environment. Numerous examples of scalability issues in past research, as recorded and discussed in Section 4.4.2, occurred due to increases by millions of features, as shown in Fig. 7. Solutions like SAOLA, as proposed by [155], have been considered one of the fastest methods due to the use of mutual information optimised by symmetrical uncertainty. SAOLA then became a benchmark method followed by other prominent researchers when developing OFS-A3M [153], OFS-Density [124], SOFS [196], and DFRS [142]. For handling relevancy and redundancy in terms of scalability in HDD, further study is required to define the relevance threshold. A solution like OFS-A3M cannot handle redundancy very effectively even though it is considered a fast method as it uses neighbourhood rough sets as the correlation measure. Meanwhile, the OFS-Density solution handles redundancy using neighbourhood rough sets and the fuzzy equal constraint, but its pitfalls are the running time and selecting the final subset of features with inconsistent training instances. DFRS can also deal with redundancy, but the consequence is high memory usage. It can be concluded that when dealing with scalability, an increase in the training size will influence evaluation measures, especially training time, accuracy, and memory allocation. The explanation of the evaluation metrics in Section 4.5 provides useful insights into dealing with redundancy, accuracy, and running time in HDD. To address the second issue, stability, OSFSMI [137] was a proposed solution that used information theory for the relevancy and redundancy analysis. This was claimed to be one of the highest-ranking methods in terms of accuracy, running time, and stability. Nevertheless, concerns remain regarding the local optima [14] problem, which requires in-depth analysis in future research.

The next issue related to HDD is the computational cost. A key purpose of developing algorithms was to decrease the computational cost of the OFS methods. SAOLA and ACO-SU [135] enhanced the algorithm by using symmetrical uncertainty to enhance information theory as a correlation measure. Using the appropriate tools can also significantly reduce the computational cost. TheSOFS and MOANOFs solutions [42] used LIBOL to deal with HDD and ultra-HDD. As Section 4.6 showed in relation to the analysis of tool performance, LIBOL appeared more able to deal with HDD than the OFS tools of LOFS and MOA. Optimisation methods like SOFS and LFBA [160] are alternative approaches to resolving high computational costs. Hence, when dealing with

the computational cost in HDD, the algorithm development, correlation measures, and tools to use should be investigated in future research. Finally, distributed processing in dynamic environments is also a critical issue. Under dynamic and streamwise circumstances, distributed parallel processing is required. Solutions such as OFFS-BP [197], OFS-ABA [133], and OSMWFS [159] were proposed to accommodate the drawback of centralised on-line processing by using distributed parallel processing. OFFS-BP utilised the Spark platform while OFS-ABA employed MapReduce to solve the issues of memory consumption and high computational cost. Meanwhile, OSMWFS proposed parallel OFS using GPU-CUDA by accessing the parallel memory of GPU and sliding windows to handle the feature redundancy and irrelevancy issue while simultaneously accelerating the processing speed. Nevertheless, various other aspects of distributed processing in dynamic environments require further exploration in the future, such as online and real-time learning, dynamic windowing size adjustment and the evolution of HDD into new dimensions.

6. Conclusion

Throughout this rigorous and systematic scrutinisation of the literature, several gaps have been identified in terms of OFS and how it has been utilised in the context of OSH data to reduce the complexity of real-world datasets. This study has also established how OFS considerably improves the learning process by selecting highly relevant subsets and eliminating irrelevant and redundant features. This paper provides the taxonomy of the OFS challenges by presenting a comprehensive review of the state-of-the-art OFS methods and the benchmark methods. Next, a framework was derived to classify the state-of-the-art methods according to OSH data behaviours. The framework categorises eight behaviours of data as the important attributes of OFS. They are group stream, high-dimensional, multi-label, class imbalance, capricious, causal-based, low redundancy with high relevancy, and feature drift. It has been emphasised that threshold initialisation, relevancy, redundancy, and dimensionality impact the performance evaluation metrics of accuracy, running time, and the number of selected features. Finally, in the framework, the tools required to develop OFS methods in different data forms have been identified, while the tools' performance with high-dimensional data has been analysed. Furthermore, to strengthen the framework, an evaluation of the capabilities of various tools was presented for different data behaviours. The discussions explicitly contextualise the framework's added value in shaping the future directions of OFS for OSH data. Nevertheless, there are a number of limitations. Firstly, the evaluation measure of the study was limited to single-label data. Second, few analyses of statistical significance were conducted and, third, no extensive experiments were undertaken in the form of relevancy and redundancy analysis. Future researchers are recommended to analyse in depth the evaluation of multi-label data and the tools' performances from different perspectives. Moreover, it would be useful to further validate the proposed framework and further enhance the modification of the OFS techniques and correlation measures, based on the findings regarding the classification of behaviours in the framework. Finally, the paper presents a comprehensive OFS classification from the perspective of data behaviours, evaluation measures, OFS tools, and the management of OFS challenges. It is hoped that the taxonomised challenges related to OFS and the derived framework will lead to extended investigations and analysis of OFS methods for OSH data in both the academic and industrial domains.

Managerial and Practical/Social Implications: This article provides insights into and knowledge of the current critical OFS issues in the context of high-dimensional data. In addition, it

has ensured that both academia and industry have an improved way to address the existing challenges by finding a suitably enhanced solution through the systematic critical review and the classification of the proposed OFS framework. The state-of-the-art OFS framework has clear theoretical and practical benefits for the community.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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